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**Немарковский Подход к Волатильности: Анализ Динамики Фьючерсов
VIX**

Path-dependent Approach to Volatility: Analysis of VIX Futures Dynamics

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Аннотация

Данная работа исследует немарковскую модель волатильности (PDV) как инструмент прогнозирования и торговли фьючерсами на VIX. Модель представляет собой привлекательного кандидата: её факторы наблюдаемы из прошлых доходностей, она допускает SDE в явной форме, хорошо калибруется совместно на SPX–VIX, а её динамика удивительно элегантна. Работа построена вокруг проверки этой гипотезы.

Разделы 1–4 кратко излагают литературу и теорию, необходимые для понимания VIX и связанных с ним инструментов. Раздел 5 определяет модель PDV и содержит теоретический результат: ядро TSPL допускает марковское представление без аппроксимации двумя экспонентами, как континуум факторов Орнштейна–Уленбека. Из этого вытекают два следствия. Двухэкспоненциальная форма Гуйона является структурной проекцией этой системы: уменьшением по размерности, а не по структуре. Кроме того, коэффициент vol-of-vol становится явным в параметрах TSPL: горизонт внимания δ_1 и скорость затухания внимания α_1 , что даёт ему ясную интерпретацию.

Раздел 6 диагностирует недостатки модели на внутрисдневных данных и мотивирует улучшения, перенесённые в прогнозный слой. Разделы 7 и 8 строят его: SDE-бенчмарк и модель CatBoost, обе протестированные при реалистичном исполнении. SDE-бенчмарк не обладает предсказательной силой: не предсказывает ни знак, ни величину. Модель CatBoost восстанавливает слабое, свободное от утечек направленное преимущество на субминутном масштабе, однако оно исходит из стакана заявок, а не из дрейфов PDV, и подвержено неблагоприятному отбору на торговом слое. Разделы 9–10 резюмируют результаты данной работы, а также рассматривают варианты, где прогнозирование и исполнение могут быть улучшены.

Abstract

This thesis explores whether Julien Guyon’s path-dependent volatility (PDV) model can be repurposed for forecasting and trading VIX futures. The model is an appealing candidate: its factors are observable from past returns, it admits a closed-form SDE, it fits SPX–VIX jointly with high accuracy, and its dynamics are surprisingly elegant. The work is organised around testing this theory.

Sections 1–4 set up the literature and theory needed for the VIX and VIX-linked instruments. Section 5 defines the PDV model and contributes a theoretical result: the TSPL kernel admits markovization, without the two-exponential approximation, as a continuum of Ornstein–Uhlenbeck factors. Two things follow. Guyon’s two-exponential form is a structural projection of that system, a truncation in dimension and not in structure. Additionally, the vol-of-vol coefficient becomes explicit in TSPL parameters: the attention horizon δ_1 and speed of memory α_1 , giving it a clear interpretation.

Section 6 diagnoses the model’s shortcomings on intraday data and motivates the fixes carried into the forecasting layer. Sections 7 and 8 build it: an SDE benchmark and a CatBoost model both tested under realistic execution. The SDE benchmark has no skill, forecasting neither sign nor magnitude. The CatBoost model recovers a faint, leakage-free directional edge at the sub-minute scale, but it comes from the order book, not the PDV drifts, and is adversely selected on the trading layer. Sections 9–10 summarise these findings by considering options where forecasting and execution can be improved.

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Introduction

In 2023, Julien Guyon and Jordan Lekeufack published “*Volatility is (mostly) path-dependent*” paper. It contained a relatively simple model that achieved $R^2 > 0.85$ in joint SPX–VIX calibration. Acknowledging its elegance and effectiveness, Risk.net named Guyon its Quant of the Year for 2025.

By design, the model is a calibration tool rather than a predictive one. It operates on daily fits optimized for mean accuracy rather than directional signal, and its inherent one-day lag makes it poorly suited for forecasting applications. Yet, it has surprisingly elegant dynamics and, on simulations, yields results similar to real life observations.

The purpose of this thesis is to explore whether the Guyon–Lekeufack path-dependent volatility model can be re-framed as a short-term forecasting tool for VIX Futures and applied for trading. Using CBOE intraday data on SPX and VX provided by FirstRateData and DataBento, we adopt a two-stage approach: a SDE benchmark of original model, followed by a Catboost approach paired with microstructure features.

This research was motivated by the author’s internship at Sber CIB (summer 2025), where the PDV model was researched with applications to the Russian market. Working on a securities trading desk where P&L generation is the primary objective, the model’s one-day lag proved problematic and the need for an intraday-ready reformulation became clear.

1 Literature review

Volatility

The implied volatility skew that appeared after the October 1987 crash falsified the assumption of constant volatility introduced by the Black–Scholes model^[1]. Its underlying follows a geometric Brownian motion with constant σ :

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

Researchers then turned to finding volatility models that both match the observed dynamics and limit arbitrage.

The local volatility model^[2] is defined as a deterministic function to match the whole European-option surface. It is recovered from call prices $C(T, K)$ by the Dupire formula

$$\sigma(T, K) = \sqrt{2 \frac{\partial_T C(T, K) + r_T K \partial_K C(T, K)}{K^2 \partial_{KK} C(T, K)}}$$

read off at $(T, K) = (t, S)$. It is arbitrage-free by construction and carries zero static error, but its dynamics are wrong: as spot moves, the local-volatility smile shifts opposite to the market^[3], which biases the delta and vega hedges.

Stochastic-volatility models add a separate variance factor correlated with spot. Heston^[4] takes

$$\begin{aligned} dF_t &= \sigma_t F_t dW_t^{(1)} \\ d\sigma_t^2 &= \kappa(\theta - \sigma_t^2) dt + \nu \sigma_t dW_t^{(2)} \\ dW_t^{(1)} dW_t^{(2)} &= \rho dt \end{aligned}$$

with κ the mean-reversion speed, θ the long-run variance, ν the volatility of volatility, and ρ the spot–volatility correlation. SABR^[3] reads

$$\begin{aligned} dF_t &= \alpha_t F_t^\beta dW_t^{(1)} \\ d\alpha_t &= \nu \alpha_t dW_t^{(2)} \\ dW_t^{(1)} dW_t^{(2)} &= \rho dt \end{aligned}$$

with β the elasticity and ν the volatility of volatility. The smile co-moves with the underlying and the dynamics are correct. However, with few parameters they cannot fit the whole surface, which leaves a permanent static error. Furthermore, volatility is an exogenous factor derived from a Brownian motion driver.

Engle’s ARCH^[5] model instead keeps volatility path-dependent. The return is the conditional volatility multiplied by the innovation term,

$$r_t = \sigma_t z_t, \quad z_t \sim \mathcal{N}(0, 1)$$

and the conditional variance is fed by past squared returns,

$$\sigma_t^2 = \omega + \sum_{i=1}^q \alpha_i r_{t-i}^2$$

GARCH^[6] generalizes it with past variances,

$$\sigma_t^2 = \omega + \sum_{i=1}^q \alpha_i r_{t-i}^2 + \sum_{i=1}^p \beta_i \sigma_{t-i}^2$$

Both are fed only by observable returns, with no latent factor to calibrate. However, they give a symmetric implied volatility smile and miss the leverage skew. Therefore, one would search for a volatility model that stays path-dependent with observable market features, restores the leverage skew GARCH lacks, remains endogenous, and recovers both the static fit and the correct dynamics.

The thesis is built around the path-dependent volatility (PDV) model of Guyon and Lekeufack^[7]. Past index returns explain up to 90% of the variance of implied volatility and 65% of future daily realized volatility. The volatility benchmark is given by

$$\begin{aligned} \sigma_t &= \beta_0 + \beta_1 R_{1,t} + \beta_2 \sqrt{R_{2,t}}, & \frac{dS_t}{S_t} &= \sigma_t dW_t \\ R_{1,t} &= \int_{-\infty}^t K_1(t-u) \frac{dS_u}{S_u}, & R_{2,t} &= \int_{-\infty}^t K_2(t-u) \left(\frac{dS_u}{S_u} \right)^2 \\ K_i(\tau) &= (\tau + \delta_i)^{-\alpha_i}, & \int_{-\infty}^t Z_{\alpha_i, \delta_i}^{-1} \cdot K_i(t-u) du &= 1 \end{aligned}$$

where R_1 weights past returns, R_2 weights past squared returns, both under time-shifted power-law kernels for short and long memory normalized to unit weight. A mixture-of-exponentials approximation of the kernels yields a 4-factor Markovian model that reproduces volatility clustering, the leverage effect, rough-like paths, and the Zumbach effect, and jointly fits SPX and VIX smiles. The same path-dependence reproduces the apparent roughness of volatility, an empirical Hurst exponent near 0.15, without any fractional driver.

Nutz and Riveros Valdevenito^[8] prove the coupled SDE system has a unique strong non-explosive solution, the volatility stays positive, and the price is a martingale. The conditional expectations in σ_t^2 have no closed form; Gazzani and Guyon^[9] calibrate via a pathwise neural approximation of the VIX that exploits the model's Markovianity. Guyon and Parent^[10] give a discrete-time 4-factor version with mixture-of-Gaussians innovations, calibrated jointly under $\mathbb{P}$ and $\mathbb{Q}$. Muravlev^[11] grounds the kernel approximation: fractional Brownian motion is a linear functional of an infinite-dimensional Ornstein–Uhlenbeck process. Wen et al.^[12] report the empirical analogue for implied volatility: a linear model with L^2 regularization explains about 80% of SPX IV variance, on par with XGBoost. Taleb^[13] argues returns have fat tails, where the finite fourth moment that GARCH estimation needs may not exist. The PDV model produces heavy tails endogenously, with no assumption on the innovations.

Market efficiency and return predictability

Forecasting VIX futures presumes returns are predictable, which the strong Efficient Market Hypothesis denies. Grossman and Stiglitz^[14] show informationally efficient markets are impossible when information is costly: if prices revealed everything, no one would pay to be informed, and prices would carry no information. So prices must stay noisy. Paying for information only pays off if the price does not already reveal it. That leftover noise is the room a forecast can use. Winter^[15] derives momentum and reversal from heterogeneous signal quality in a rational expectations equilibrium, with no behavioral biases. Andrei and Cujean^[16] obtain short-term momentum and long-term reversal jointly from the distribution of information. Predictability follows from information frictions, not irrationality. Past returns are a slow, partly public signal, so a path-dependent forecast lives exactly in the friction these models describe.

Market microstructure

The thesis compares this parametric model with a CatBoost forecaster built on order-book features. The order book is useful for both forecasting and trading. Harris^[17] is the practitioner reference on how orders, traders, and exchange mechanics form prices. Gould et al.^[18] survey limit-order-book modeling. Kyle^[19] measures price impact with a single number, λ , and shows that the price moves linearly in the net order flow v

$$p = p_0 + \lambda v.$$

A bigger λ means an order moves the price more. It rises when order flow may be informed. Avellaneda and Stoikov^[20] introduce the canonical market-making model that quotes around an inventory-adjusted reservation price

$$r = s \pm q\gamma\sigma^2(T - t),$$

where s is the mid-price, q the inventory, γ the risk aversion, σ the volatility, $T - t$ the time to horizon. This elegant theory breaks in real life, and Dixon^[21] shows how the order book can be used for market making and forecasting: he casts short-horizon price prediction from order-book features as supervised learning. This is the template for the CatBoost stage.

Machine learning and filtering

Dixon, Halperin and Bilokon^[22] is the methodological reference for the machine-learning models. The thesis borrows their out-of-sample protocol: the training, validation, and testing split and the time-series cross-validation, used here as the walk-forward scheme. Tree depth and early stopping are tuned on the validation split. The forecasts are scored with their classification metrics. The thesis uses boosted trees. They capture non-linear feature interactions with little preprocessing. Javaheri, Lautier and Galli^[23] cover the Kalman filter and its nonlinear variants for extracting a signal from noisy observations.

2 Theoretical Foundations

What is the VIX, and how is it computed? Informally, it is the average of expected future volatility over a fixed horizon. Formally, this section gives the strict derivation, starting from general theory on volatility instruments.

2.1 Forward Variance

For theory consider model of the form:

$$dS_t = r_t S_t dt + \sigma_t S_t dW_t \quad (1)$$

where r_t is non-random and also assume $\mathbb{E} \int_0^T \sigma_t^2 dt < \infty$.

Definition 2.1. The T -forward variance at time t is defined by

$$\xi_t^T = \mathbb{E}(\sigma_T^2 \mid \mathcal{F}_t), \quad t \in [0, T]. \quad (2)$$

The forward variance curve at time t is the function $T \mapsto \xi_t^T$, $T \in [t, \infty)$. Furthermore, ξ_t^T is $\mathcal{F}_t$ -measurable, $\xi_t^t = \sigma_t^2$ and $\xi_t^T = (\xi_t^T)_{t \leq T}$ is a martingale.

Example: Forward Variance in the Heston model.

Recall the Heston model:

$$dS_t = r_t S_t dt + \sqrt{V_t} dW_t^{(1)}, \quad (3)$$

$$dV_t = \kappa(\theta - V_t) dt + \sigma \sqrt{V_t} dW_t^{(2)}. \quad (4)$$

The forward variance curve in the Heston model is

$$\xi_t^T = \theta + e^{-\kappa(T-t)} (\xi_t^t - \theta), \quad (5)$$

and it satisfies the SDE

$$d\xi_t^T = \sigma e^{-\kappa(T-t)} \sqrt{\xi_t^t} dW_t^{(2)}, \quad t \leq T. \quad (6)$$

Note that $\xi_t^t = V_t$. If $V_t = \theta$, then the forward variance curve is flat ($\xi_t^T = \theta$ for all $T \geq t$).

Proof. From the model equations, we have

$$V_T = V_t + \kappa \int_t^T (\theta - V_s) ds + \sigma \int_t^T \sqrt{V_s} dW_s^{(2)}. \quad (7)$$

Taking the conditional expectation $\mathbb{E}(\cdot \mid \mathcal{F}_t)$, we get

$$\xi_t^T = V_t + \kappa \int_t^T (\theta - \xi_t^s) ds. \quad (8)$$

Differentiating:

$$\frac{\partial}{\partial T} \xi_t^T = \kappa(\theta - \xi_t^T). \quad (9)$$

The solution of this equation is given by the formula in the proposition. The SDE follows by Itô's formula. $\square$

2.2 Variance Swaps and Forward Variance

Definition 2.2. The *realized variance* of the stock price S in an interval $[t, T]$, where $T - t = n\Delta t$, is defined by

$$\text{RV}_t^T(S) = \frac{1}{T-t} \sum_{i=1}^n \left(\ln \frac{S_{t+i\Delta t}}{S_{t+(i-1)\Delta t}} \right)^2. \quad (10)$$

In practice, time is measured in years, $\Delta t \approx 1/250$ (one trading day) and n is the number of days in the period $[t, T]$.

Definition 2.3. A *variance swap* (VS) is a contract to exchange the realized variance of the stock price for a fixed cash amount. Its payoff is given by

$$X = \text{RV}_t^T(S) - (\hat{\sigma}_t^T)^2. \quad (11)$$

The quantity $\hat{\sigma}_t^T$ is set such that the value of the payoff at time t is zero and is called the *VS volatility* for maturity T and $(\hat{\sigma}_t^T)^2$ is called the *VS variance*.

Proposition 2.4. Assume that the process S_t satisfies the SDE

$$dS_t = \mu_t S_t dt + \sigma_t S_t dW_t. \quad (*)$$

Let $\Delta t = T/n$. Then

$$\sum_{i=1}^n \left(\ln \frac{S_{t+i\Delta t}}{S_{t+(i-1)\Delta t}} \right)^2 \xrightarrow{P} \int_t^T \sigma_s^2 ds \quad \text{as } n \rightarrow \infty. \quad (12)$$

Proof. As $n \rightarrow \infty$, the log-increments satisfy $\ln \frac{S_{t+i\Delta t}}{S_{t+(i-1)\Delta t}} \rightarrow d \ln S_{t+(i-1)\Delta t}$. By Itô's formula applied to $\ln S_t$,

$$d \ln S_t = \left(\mu_t - \frac{\sigma_t^2}{2} \right) dt + \sigma_t dW_t, \quad (13)$$

so $(d \ln S_t)^2 = \sigma_t^2 dt$. Therefore

$$\sum_{i=1}^n \left(\ln \frac{S_{t+i\Delta t}}{S_{t+(i-1)\Delta t}} \right)^2 \xrightarrow{P} [\ln S]_t^T = \int_t^T (d \ln S_u)^2 = \int_t^T \sigma_s^2 ds. \quad (14)$$

□

Definition 2.5. The *variance swap variance* for maturity T at time $t \leq T$ is defined by

$$(\hat{\sigma}_t^T)^2 = \frac{1}{T-t} \mathbb{E} \left(\int_t^T \sigma_s^2 ds \mid \mathcal{F}_t \right). \quad (15)$$

This is the value of $(\hat{\sigma}_t^T)^2$ that sets the present value of payoff (11) to zero at time t . Indeed, replacing $\text{RV}_t^T(S)$ by its continuous limit from the preceding proposition,

$$\mathbb{E}(X \mid \mathcal{F}_t) = \frac{1}{T-t} \mathbb{E} \left(\int_t^T \sigma_s^2 ds \mid \mathcal{F}_t \right) - (\hat{\sigma}_t^T)^2 = 0. \quad (16)$$

2.3 Replication

Proposition 2.6. *We have*

$$\int_t^T \sigma_u^2 \, du = 2 \int_t^T \frac{dS_u}{S_u} - 2 \ln \frac{S_T}{S_t}. \quad (17)$$

Proof. By Itô's formula, $d \ln S_t = \frac{dS_t}{S_t} - \frac{1}{2} \sigma_t^2 \, dt$, so

$$\sigma_t^2 \, dt = 2 \frac{dS_t}{S_t} - 2 \, d \ln S_t. \quad (18)$$

Integrating from t to T gives the result. $\square$

Remark 2.7. The replication has two legs. $\int_t^T \frac{dS_u}{S_u}$ is the P&L of continuously holding $\frac{1}{S_u}$ shares in stock – always \$1 invested, nothing more. The term $-2 \ln \frac{S_T}{S_t}$ is the payoff of a log-contract, bought once at t and held to T .

Remark 2.8. Keeping \$1 in stock does not freeze the portfolio – the stock moves and P&L accrues. What is constant is the notional: after each move you reset to $\frac{1}{S_u}$ shares, so the dollar exposure is always \$1. The integral accumulates all these infinitesimal gains.

Proposition 2.9 (Taylor replication).

Let $g : (0, \infty) \rightarrow \mathbb{R}$ be twice continuously differentiable and let $x > 0$. Then for all $y > 0$,

$$g(y) = g(x) + g'(x)(y - x) + \int_0^x (K - y)^+ g''(K) \, dK + \int_x^\infty (y - K)^+ g''(K) \, dK. \quad (19)$$

Proof. By the fundamental theorem of calculus and integration by parts with $v = s - y$,

$$\begin{aligned} g(y) - g(x) &= \int_x^y g'(s) \, ds = [g'(s)(s - y)]_x^y - \int_x^y g''(s)(s - y) \, ds \\ &= g'(x)(y - x) + \int_x^y (y - s) g''(s) \, ds. \end{aligned} \quad (20)$$

For $y > x$: $\int_x^y (y - s) g''(s) \, ds = \int_x^\infty (y - K)^+ g''(K) \, dK$ and $\int_0^x (K - y)^+ g''(K) \, dK = 0$. For $y < x$ the argument is symmetric with puts and calls swapped. Combining both cases gives the result. $\square$

Application: log decomposition.

Setting $g(y) = -\ln(y/x)$, so $g(x) = 0$, $g'(x) = -1/x$, $g''(K) = 1/K^2$:

$$-\ln \frac{S_T}{x} = -\frac{S_T - x}{x} + \int_0^x \frac{(K - S_T)^+}{K^2} \, dK + \int_x^\infty \frac{(S_T - K)^+}{K^2} \, dK. \quad (21)$$

Proposition 2.10 (Valuation). *Let $\hat{C}_t^T(K) = \mathbb{E}((S_T - K)^+ | \mathcal{F}_t)$ and $\hat{P}_t^T(K) = \mathbb{E}((K - S_T)^+ | \mathcal{F}_t)$ denote undiscounted call and put prices with maturity T and strike K at time t , and let $F_t^T = \mathbb{E}(S_T | \mathcal{F}_t)$ be the T -forward price. Then*

$$(\hat{\sigma}_t^T)^2 = \frac{2}{T-t} \int_0^{F_t^T} \frac{\hat{P}_t^T(K)}{K^2} dK + \frac{2}{T-t} \int_{F_t^T}^{\infty} \frac{\hat{C}_t^T(K)}{K^2} dK. \quad (22)$$

Proof. Step 1. Since $dF_t^T = \sigma_t F_t^T dW_t$, Itô's formula gives

$$\ln F_T^T = \ln F_t^T + \int_t^T \sigma_s dW_s - \frac{1}{2} \int_t^T \sigma_s^2 ds. \quad (23)$$

Taking $\mathbb{E}(\cdot | \mathcal{F}_t)$, using $F_T^T = S_T$ and the definition of $(\hat{\sigma}_t^T)^2$,

$$(\hat{\sigma}_t^T)^2 = -\frac{2}{T-t} \mathbb{E} \left(\ln \frac{S_T}{F_t^T} \middle| \mathcal{F}_t \right). \quad (24)$$

Step 2. Apply the log decomposition with $x = F_t^T$ and take $\mathbb{E}(\cdot | \mathcal{F}_t)$:

$$-\mathbb{E} \left(\ln \frac{S_T}{F_t^T} \middle| \mathcal{F}_t \right) = -\frac{\mathbb{E}(S_T | \mathcal{F}_t) - F_t^T}{F_t^T} + \int_0^{F_t^T} \frac{\hat{P}_t^T(K)}{K^2} dK + \int_{F_t^T}^{\infty} \frac{\hat{C}_t^T(K)}{K^2} dK. \quad (25)$$

Since $\mathbb{E}(S_T | \mathcal{F}_t) = F_t^T$, the first term vanishes. Multiplying by $\frac{2}{T-t}$ gives the result. $\square$

3 The VIX Index

The VIX index is the volatility index of the S&P 500, defined as

$$\text{VIX}_t = \hat{\sigma}_t^{t+\tau}, \quad (26)$$

where $\tau = 30/365$ (30 calendar days).

Using the definition of the VS volatility from Section 2,

$$\text{VIX}_t = \sqrt{\frac{1}{\tau} \mathbb{E} \left(\int_t^{t+\tau} \sigma_s^2 ds \middle| \mathcal{F}_t \right)} = \sqrt{\frac{1}{\tau} \int_t^{t+\tau} \xi_t^s ds}, \quad (27)$$

where $\xi_t^s = \mathbb{E}(\sigma_s^2 | \mathcal{F}_t)$ is the forward variance from Definition 2.1.

Example

If one sees VIX quoted at 20 right now, then the expected integrated variance over the next 30 days, annualized and square-rooted, equals 0.20.

In practice, the VIX is computed from market prices of SPX options using the result of Proposition 2.10, which expresses the VS variance through call and put prices.

3.1 The CBOE Calculation

The algorithm used by CBOE discretizes the continuous option-price integral. The generalized formula is:

$$\sigma^2 = \frac{2}{T} \sum_i \frac{\Delta K_i}{K_i^2} e^{rT} Q(K_i) - \frac{1}{T} \left[\frac{F}{K_0} - 1 \right]^2, \quad (28)$$

where:

- T – time to expiration (in years),
- F – option-implied forward price,
- K_0 – first strike equal to or immediately below F ,
- K_i – strike price of the i -th out-of-the-money option: a call if $K_i > K_0$, a put if $K_i < K_0$; both put and call if $K_i = K_0$,
- $\Delta K_i = (K_{i+1} - K_{i-1})/2$ – half the distance between adjacent strikes,
- $Q(K_i)$ – midpoint of the bid–ask spread for the option with strike K_i ,
- r – risk-free interest rate to expiration.

The second term corrects for the fact that $K_0 \neq F$ in general due to discrete net of strikes. The VIX index is then

$$\text{VIX} = 100 \times \sigma. \quad (29)$$

Remark 3.1.

1. To target exactly 30 calendar days, CBOE interpolates between two expiration dates T_1 and T_2 bracketing τ (with $T_1 \leq \tau \leq T_2$), computing σ_1^2 and σ_2^2 from (28) separately and blending them by time weight.
2. SPX options used in the calculation are European-style. CBOE includes only options with non-zero bid prices and truncates the strike range after two consecutive zero-bid strikes.
3. The VIX is a *derived index*, not a traded asset. One cannot buy or sell “the VIX” itself. It is a number computed in real time from SPX option prices and published by CBOE.

4 The VIX Futures

A VIX future with maturity T is an exchange-traded contract whose settlement price at expiration equals VIX_T – the value of the VIX index on the expiration date. But its price *before* expiration is not determined by the current VIX spot, but by current expectations of the VIX index at maturity T .

4.1 Pricing

For equity futures, the fair value is given by cost-of-carry: $F = S e^{r(T-t)}$. This works because one can buy and hold the underlying stock. The VIX is not a traded asset, so there is no position one can take in “the VIX spot”, and so cost-of-carry does not apply.

Instead, the fair value of a VIX future is

$$F_t^{\text{VIX},T} = \mathbb{E}^{\mathbb{Q}}(\text{VIX}_T \mid \mathcal{F}_t), \quad (30)$$

the risk-neutral conditional expectation of the VIX at maturity. This expectation depends on the entire volatility-of-volatility surface and cannot in general be expressed in closed form.

4.2 What the Quote Shows

Suppose on April 1, 2026, one observes:

- VIX spot: 18.5
- VIX May future (expires May 20): 20.2

The spot 18.5 is the market’s current estimate of 30-day S&P 500 volatility, computed from today’s SPX options via (28). The May future quote 20.2 is the market’s current estimate of what the VIX will be on May 20 – i.e., what SPX option-implied 30-day volatility will be on that date. The settlement price of the May future will be whatever VIX actually reads on May 20.

The current VIX futures quote is a forecast of future implied volatility, not current implied volatility.

4.3 Term Structure

The curve $T \mapsto F_t^{\text{VIX},T}$ is the VIX futures term structure. It typically exhibits *contango*: longer-dated futures trade above the spot VIX. This reflects mean reversion of volatility – when VIX is below its long-run average (≈ 20), the market prices future volatility higher than current. During market stress, the term structure can invert (*backwardation*): near-term futures exceed far-dated ones because the market expects elevated volatility to subside.

4.4 How one would trade VIX Futures

1. Suppose you are a trader and see that the VIX futures contract expiring on June 30 is quoted at 20. This means the market's current consensus is that the VIX index will read 20 on June 30. You believe volatility will be higher – perhaps due to upcoming elections or macro uncertainty – so you enter long at 20.
2. Over the following weeks, the market revises its expectation upward and the futures price rises to 22. A new trader entering long now pays 22 – they need VIX above 22 at expiration to profit. Your position, entered at 20, already shows an unrealized gain of 2 points.
3. On June 30, the VIX index is computed from SPX options and reads 25. This is the *settlement price* – determined once, at expiration, by the actual VIX value. It is not fixed in advance. Your P&L is $25 - 20 = 5$ points. The trader who entered at 22 makes $25 - 22 = 3$ points. Had VIX settled at 18, you would have lost 2 points.
4. The key point: a VIX futures contract is a bet on where the VIX index will be at expiration. There is no fixed payout. The futures price adjusts continuously as the market's estimate of future implied volatility changes.

5 The Guyon–Lekeufack Volatility Model

In 2023, Julien Guyon and Jordan Lekeufack^[7] introduced a relatively simple model of the past returns that captures roughly 85% of the out-of-sample variation in the VIX and about 65% of realized volatility. Through it, they solve the puzzle of joint SPX-VIX implied volatility calibration. Furthermore, their model kills five birds with one stone by matching stylized facts about volatility and can be nicely Markovised.

5.1 Background

There are 3 volatility model classes in general. We don't go deep into the details here, but each has a known limitation. Here, in simple, we judge them on two criteria: static error produced by model, and dynamic error produced by volatility smile shift.

Stochastic volatility.

Stochastic volatility models introduce separate Brownian motion factors for volatility. Smile dynamics are correct: the surface co-moves with spot. But the surface cannot be fit exactly with few parameters – a permanent static error remains. Furthermore, these models are often exogenous, rather than endogenous in explaining volatility. Heston^[4] is the example.

Local volatility.

Local volatility is special class of models that are made to exactly set static error to 0 by specifying $\sigma_t = \sigma(t, S_t)$. But the dynamics are wrong: as spot moves, the LV-implied smile shifts opposite to what markets show (ex. Dupire's local volatility model^[2]). Hedge ratios and smile-sensitive exotic prices are biased.

Path-dependent volatility.

A third class expresses σ_t as a function of the past observations, $\sigma_t = F((S_u)_{u \leq t})$, without extra randomness. The classical example is GARCH^[6], but its theory requires a finite fourth moment of returns. Taleb^[13] argues that returns empirically have a tail index below 4, which questions the consistency of the estimators obtained.

Remark 5.1.

This is a shallow introduction to volatility pricing, aimed only at the main ideas of each approach. The author admits that different models serve different purposes and outperform one another on different tasks.

Mixtures of approaches also exist: optimal transport models, local-stochastic models and so on. In the author's opinion, Guyon's model acts as a balanced sword: it captures almost everything while staying relatively simple.

The Guyon–Lekeufack model.

The 2023 model stays path-dependent but compacts information into two features and remains linear in specification:

$$\sigma_t = \beta_0 + \beta_1 R_{1,t} + \beta_2 \sqrt{R_{2,t}}, \quad \beta_0 > 0, \beta_1 < 0, \beta_2 \in (0, 1). \quad (31)$$

where

$$R_{1,t} = \sum_{t_i \leq t} K_1(t - t_i) r_{t_i}, \quad R_{2,t} = \sum_{t_i \leq t} K_2(t - t_i) r_{t_i}^2, \quad (32)$$

are weighted moving averages of past signed and squared returns of SPX (full definitions in Section 5.3). σ_t can be any volatility metric: VIX, realized volatility, or next-day predicted volatility.

5.2 Stylized Facts of Volatility

Guyon & Lekeufack^[7] list five stylized facts the PDV model matches.

1. *Leverage effect.* Negative returns are followed by higher volatility:

$$\text{Corr}(r_t, \sigma_{t+h}) < 0, \quad h > 0. \quad (33)$$

2. *Volatility clustering.* High and low volatility periods.

3. *Volatility roughness.* Log-volatility satisfies

$$\mathbb{E} |\ln \sigma_{t+\Delta} - \ln \sigma_t|^q \sim c \Delta^{qH} \quad \text{as } \Delta \rightarrow 0, \quad (34)$$

with empirical Hurst exponent $H \approx 0.15$, well below the standard Brownian motion $H = 1/2$. This property often motivates the *rough volatility* literature.

4. *Weak Zumbach effect.* Past squared returns forecast future volatility more strongly than past volatility forecasts future squared returns:

$$\text{Corr}\left(\left(\frac{dS_t}{S_t}\right)^2, \sigma_{t+h}\right) > \text{Corr}\left(\sigma_t, \left(\frac{dS_{t+h}}{S_{t+h}}\right)^2\right), \quad h > 0. \quad (35)$$

Equivalently, the joint process (S, σ) is not invariant under time reversal. Explanatory power is lost when read backwards versus forwards in time.

5. *Strong Zumbach effect.* The conditional law $\text{Law}(\sigma_{t+h} | \mathcal{F}_t)$ depends not only on the past path of σ , but also on the past path of returns $(dS_u/S_u)_{u \leq t}$.

Guyon's PDV captures all of the above nicely without aiming to fit any of them specifically.

5.3 The Model and Its Kernel

Let t_i denote the discrete observation times (e.g. daily closes) and

$$r_{t_i} = \frac{S_{t_i} - S_{t_{i-1}}}{S_{t_{i-1}}} \quad (36)$$

the one-period return of the SPX index. Guyon & Lekeufack^[7] then propose

$$\sigma_t = \beta_0 + \beta_1 R_{1,t} + \beta_2 \sqrt{R_{2,t}}, \quad \beta_0 > 0, \quad \beta_1 < 0, \quad \beta_2 \in (0, 1), \quad (37)$$

with features

$$R_{1,t} = \sum_{t_i \leq t} K_1(t - t_i) r_{t_i}, \quad R_{2,t} = \sum_{t_i \leq t} K_2(t - t_i) r_{t_i}^2. \quad (38)$$

Term by term:

1. $\beta_0 > 0$ fixes the baseline level of volatility.
2. $R_{1,t}$ is a weighted average of past signed returns. This captures the leverage relation between volatility and returns: a recent down-move in the index raises σ_t .
3. $R_{2,t}$ is a weighted average of past squared returns, which serves as a proxy for the recent state of realised variance. The term $\beta_2 \sqrt{R_{2,t}}$ carries the magnitude of recent activity.

Furthermore, two observations:

1. The model targets σ directly rather than σ^2 , so every term is homogeneous to a volatility
2. Coefficient $\beta_2 > 0$ drives clustering through persistence of $\sqrt{R_{2,t}}$, and $\beta_2 < 1$ keeps the recursion stable. This is similar to GARCH requirements.

The kernels K_1, K_2 weight past returns according to their recency. Two properties are present:

1. Recent returns dominate, but the kernel does not blow up as $\tau \rightarrow 0^+$. This mimics the human brain, putting the highest attention on the most recent actions.
2. Decay is slow enough to carry long-range memory. This is a well-documented feature of volatility, but we advise to consult the original paper on how this was obtained.

In order to capture the long lasting memory of volatility, kernels K_1, K_2 are set to *time-shifted power laws* or *TSPL* in short:

$$K_{\alpha,\delta}(\tau) = \frac{1}{Z_{\alpha,\delta}} (\tau + \delta)^{-\alpha}, \quad Z_{\alpha,\delta} = \int_0^\infty (\tau + \delta)^{-\alpha} d\tau = \frac{\delta^{1-\alpha}}{\alpha - 1}, \quad (39)$$

1. $Z_{\alpha,\delta}$ acts as a normalising constant such that density is summed up to 1
2. $\delta > 0$ is needed to remove singularity at $\tau = 0$ and empirically needed to capture the center of "memory"
3. $\alpha > 1$ is needed to calibrate the weight of memory and in addition regulates the size of tail.
4. Each of K_1 and K_2 takes this form with its own (α_i, δ_i) , fit to data.
5. In practice the convolution of returns is truncated at a finite horizon (Guyon & Lekeufack use four business years),

Remark 5.2.

A natural alternative is the single exponential kernel $K_\lambda(\tau) = \lambda e^{-\lambda\tau}$. Guyon & Lekeufack^[7] test this form with λ selected via Lasso regression over a grid and find that large- λ (short-memory) kernels are rarely chosen. This is why the heavy-tailed TSPL is used.

5.4 Continuous-Time Dynamics

The continuous-time dynamics that can be used for Monte Carlo simulations and theoretically for short-term predictions is derived below

$$\begin{cases} dS_t/S_t = \sigma_t dW_t, \\ \sigma_t = \beta_0 + \beta_1 R_{1,t} + \beta_2 \sqrt{R_{2,t}}, \\ R_{1,t} = \int_{-\infty}^t K_1(t-u) \cdot (dS_u/S_u) = \int_{-\infty}^t K_1(t-u) \sigma_u dW_u \\ R_{2,t} = \int_{-\infty}^t K_2(t-u) \cdot (dS_u/S_u)^2 = \int_{-\infty}^t K_2(t-u) \sigma_u^2 du. \end{cases} \quad (40)$$

For a general kernel, differentiating these expressions produces terms that depend on the full history in $\mathcal{F}_t$. The system is non-Markovian. Simulation is expensive.

Yet, by cleverly choosing the kernel (hint: derivative of exponent is the exponent itself) we can markovise the model such that $\mathcal{F}_t$ is compacted into state variables.

The 2-EXP approximation.

Guyon & Lekeufack^[7] replace each TSPL kernel by a convex combination of two exponentials:

$$K_{\theta, \lambda_0, \lambda_1}(\tau) = (1 - \theta) \lambda_0 e^{-\lambda_0 \tau} + \theta \lambda_1 e^{-\lambda_1 \tau}, \quad \theta \in [0, 1], \lambda_0 > \lambda_1 > 0. \quad (41)$$

where parameters can be explained as:

1. λ_0 (large): weight on very recent returns – short memory.
2. λ_1 (small): slow decay at long lags – long memory.
3. θ : mixing between the two scales.

Empirically, this approximation is visually indistinguishable when TSPL is plotted against 2EXP.

We now derive the 4-factor Markovian model obtained by using 2EXP instead of TSPL.

The 4-factor PDV model.

For $n \in \{1, 2\}$ and $j \in \{0, 1\}$, define

$$R_{n,j,t} = \int_{-\infty}^t \lambda_{n,j} e^{-\lambda_{n,j}(t-u)} \left(\frac{dS_u}{S_u} \right)^n, \quad (42)$$

with the fact that $(dS_u/S_u)^1 = \sigma_u dW_u$ and $(dS_u/S_u)^2 = \sigma_u^2 du$.

Then $R_{n,t} = (1 - \theta_n)R_{n,0,t} + \theta_n R_{n,1,t}$ and each sub-factor satisfies a closed-form SDE:

$$\begin{cases} dR_{1,j,t} = \lambda_{1,j}(\sigma_t dW_t - R_{1,j,t} dt), \\ dR_{2,j,t} = \lambda_{2,j}(\sigma_t^2 - R_{2,j,t}) dt, \end{cases} \quad j \in \{0, 1\}. \quad (43)$$

Derivation. Non-strict. For approximation fans.

Fix n, j and set $K(\tau) = \lambda_{n,j} e^{-\lambda_{n,j}\tau}$. Split the defining integral at the upper endpoint:

$$R_{n,j,t+dt} = \int_{-\infty}^t K(t+dt-u) \left(\frac{dS_u}{S_u} \right)^n + \int_t^{t+dt} K(t+dt-u) \left(\frac{dS_u}{S_u} \right)^n.$$

1. The first integral factors as $e^{-\lambda_{n,j}dt} R_{n,j,t}$, since $K(t+dt-u) = e^{-\lambda_{n,j}dt} K(t-u)$
2. The second is approximated by its lower-endpoint value $u = t$, where $K(t+dt-u) = K(dt) = \lambda_{n,j}(1 - \lambda_{n,j}dt) \approx \lambda_{n,j}$ due to $(dS_t/S_t)^n$ being of order dt or dW .

Therefore

$$R_{n,j,t+dt} \approx e^{-\lambda_{n,j}dt} R_{n,j,t} + \lambda_{n,j} \left(\frac{dS_t}{S_t} \right)^n.$$

Subtracting $R_{n,j,t}$ and expanding $e^{-\lambda_{n,j}dt} = 1 - \lambda_{n,j}dt + O(dt^2)$ to first order,

$$dR_{n,j,t} = \lambda_{n,j} \left[\left(\frac{dS_t}{S_t} \right)^n - R_{n,j,t} dt \right].$$

Substituting $(dS_t/S_t)^n$ recovers the two SDE lines above.

Derivation. Strict. For Ito's fans.

Fix n, j and set $\lambda = \lambda_{n,j}$. Write

$$R_t = \int_{-\infty}^t \lambda e^{-\lambda(t-u)} \left(\frac{dS_u}{S_u} \right)^n = \lambda e^{-\lambda t} \underbrace{\int_{-\infty}^t e^{\lambda u} \left(\frac{dS_u}{S_u} \right)^n}_{I_t}.$$

By Ito's product rule on $R_t = \lambda e^{-\lambda t} I_t$, with $dI_t = e^{\lambda t} (dS_t/S_t)^n$, we obtain

$$dR_t = -\lambda R_t dt + \lambda e^{-\lambda t} dI_t = \lambda \left[\left(\frac{dS_t}{S_t} \right)^n - R_t dt \right].$$

Substituting $(dS_t/S_t)^1 = \sigma_t dW_t$ and $(dS_t/S_t)^2 = \sigma_t^2 dt$ recovers the two SDEs.

Dynamics of the instantaneous volatility.

Aggregate the sub-factor SDEs for $n = 1$,

$$\begin{aligned} dR_{1,t} &= (1 - \theta_1) dR_{1,0,t} + \theta_1 dR_{1,1,t} = \\ &= [(1 - \theta_1)\lambda_{1,0} + \theta_1\lambda_{1,1}] \sigma_t dW_t - [(1 - \theta_1)\lambda_{1,0}R_{1,0,t} + \theta_1\lambda_{1,1}R_{1,1,t}] dt. \end{aligned}$$

The bracketed coefficients motivate the weighted averages

$$\bar{\lambda}_n := (1 - \theta_n)\lambda_{n,0} + \theta_n\lambda_{n,1}, \quad \bar{R}_{n,t} := \frac{(1 - \theta_n)\lambda_{n,0}R_{n,0,t} + \theta_n\lambda_{n,1}R_{n,1,t}}{\bar{\lambda}_n}, \quad (44)$$

in terms of which

$$dR_{1,t} = \bar{\lambda}_1(\sigma_t dW_t - \bar{R}_{1,t} dt), \quad dR_{2,t} = \bar{\lambda}_2(\sigma_t^2 - \bar{R}_{2,t}) dt,$$

where $dR_{2,t}$ follows by the same aggregation argument for $n = 2$.

Since $R_{2,t}$ has no dW component, Ito's formula gives $d\sqrt{R_{2,t}} = dR_{2,t}/(2\sqrt{R_{2,t}})$.

Differentiating $\sigma_t = \beta_0 + \beta_1 R_{1,t} + \beta_2 \sqrt{R_{2,t}}$ yields

$$d\sigma_t = \left(-\beta_1 \bar{\lambda}_1 \bar{R}_{1,t} + \frac{\beta_2 \bar{\lambda}_2}{2} \frac{\sigma_t^2 - \bar{R}_{2,t}}{\sqrt{R_{2,t}}} \right) dt + \beta_1 \bar{\lambda}_1 \sigma_t dW_t. \quad (45)$$

Term by term:

- *Leverage effect.* The term $\beta_1 \bar{\lambda}_1 \sigma_t dW_t$ ties σ_t to the return $\sigma_t dW_t = dS/S$; with $\beta_1 < 0$ a down-move raises volatility.
- *Mean reversion effect.* The term $-\beta_1 \bar{\lambda}_1 \bar{R}_{1,t}$ pushes σ_t downward when recent returns are negative.
- *Volatility clustering.* The term $\frac{\beta_2 \bar{\lambda}_2}{2} (\sigma_t^2 - \bar{R}_{2,t})/\sqrt{R_{2,t}}$ pulls σ_t toward its path-weighted mean variance.

Remark 5.3. Fast mean reversion resolves the positive-VIX-skew puzzle. Fitted $\lambda_{n,j}$ are large, which keeps σ_t close to its average (low) level but still allows high jumps that generate positive skew.

Remark 5.4. Although the model's paths are generated by Brownian motion ($H = 1/2$), daily sampling of σ_t yields $\hat{H} \approx 0.15$. The kernel-weighted approach produces variation that the Hurst estimator cannot distinguish from fractional roughness.

Remark 5.5.

Each $R_{n,j,t}$ equation is an Ornstein–Uhlenbeck process mean-reverting towards a moving signal. $R_{2,j,t}$ reverts to σ_t^2 at rate $\lambda_{2,j}$; $R_{1,j,t}$ reverts to 0 while being driven by $\sigma_t dW_t$.

To sum up, this yields the following Markovian version of the model:

$$\frac{dS_t}{S_t} = \sigma_t dW_t \quad (46)$$

$$\sigma_t = \sigma(R_{1,t}, R_{2,t}) \quad (47)$$

$$\sigma(R_1, R_2) = \beta_0 + \beta_1 R_1 + \beta_2 \sqrt{R_2} \quad (48)$$

$$R_{n,j,t} := \int_{-\infty}^t \lambda_{n,j} e^{-\lambda_{n,j}(t-u)} \left(\frac{dS_u}{S_u} \right)^n \quad (49)$$

$$R_{1,t} = (1 - \theta_1) R_{1,0,t} + \theta_1 R_{1,1,t} \quad (50)$$

$$R_{2,t} = (1 - \theta_2) R_{2,0,t} + \theta_2 R_{2,1,t} \quad (51)$$

$$\begin{aligned} dR_{1,j,t} &= \lambda_{1,j} \left(\frac{dS_t}{S_t} - R_{1,j,t} dt \right) = \\ &= \lambda_{1,j} (\sigma(R_{1,t}, R_{2,t}) dW_t - R_{1,j,t} dt), \quad j \in \{0, 1\} \end{aligned} \quad (52)$$

$$\begin{aligned} dR_{2,j,t} &= \lambda_{2,j} \left(\left(\frac{dS_t}{S_t} \right)^2 - R_{2,j,t} dt \right) = \\ &= \lambda_{2,j} (\sigma(R_{1,t}, R_{2,t})^2 - R_{2,j,t} dt), \quad j \in \{0, 1\}. \end{aligned} \quad (53)$$

$$d\sigma_t = \left(-\beta_1 \bar{\lambda}_1 \bar{R}_{1,t} + \frac{\beta_2 \bar{\lambda}_2}{2} \frac{\sigma_t^2 - \bar{R}_{2,t}}{\sqrt{R_{2,t}}} \right) dt + \beta_1 \bar{\lambda}_1 \sigma_t dW_t. \quad (54)$$

5.5 Wellposedness

This section states the wellposedness properties of the model. First a brief recap of what these properties mean.

Definition 5.6. *Weak existence*

Weak existence guarantees that there exists a filtered probability space and a pair (X, W) satisfying the SDE. In short, the entire law of X matches the SDE, which allows for path-dependent pricing.

Definition 5.7. *Strong existence*

Strong existence claims that on the same probability space we can construct X as a functional of W , i.e. $X = \Phi(W)$. This allows for pathwise operations: greeking, antithetic-variable variance reduction, etc.

Definition 5.8. *Weak uniqueness*

Weak uniqueness states that any two weak solutions, regardless of implementation (for simplicity, say Euler or Milstein), yield the same law for X . This ensures that the prices obtained are the same across implementations.

Definition 5.9. *Strong uniqueness*

Strong uniqueness ensures that the mapping $X = \Phi(W)$ is unique. If two different libraries both claim to simulate the model correctly, they must produce identical paths when fed the same random seed.

Properties [8]

1. Under the conditions $\beta_1 \leq 0$; $\beta_0, \beta_2, \lambda_{n,j} \geq 0$; $\theta_1, \theta_2 \in [0, 1]$, the 4-factor SDE has a strong, unique solution.
In other words, simulation is reproducible and stable, and pathwise operations are valid.
2. In the 2-factor model obtained by setting $\theta_1 = \theta_2 = 0$, if the initial volatility $\sigma_0 > 0$ and $\lambda_2 < 2\lambda_1$, then $\sigma_t > 0$ for all $t \geq 0$ a.s.

The condition $\lambda_2 < 2\lambda_1$ is satisfied in the original paper. In the 4-factor case, strict positivity is not guaranteed, but holds empirically.

3. In the 2-factor model, the process X_t satisfying $dX_t = \sigma_t X_t dW_t$ is a true martingale, not only a local martingale.

In other words, standard risk-neutral pricing applies without adjustment.

5.6 Personal theory add-on

Before moving on to practical implementation of this project's goal, we show that system (46)–(54) can be reached without approximation of the TSPL kernel by 2EXP.

We start from the non-Markovian PDV system with TSPL kernels:

$$\frac{dS_t}{S_t} = \sigma_t dW_t, \quad (55)$$

$$\sigma_t = \sigma(R_{1,t}, R_{2,t}), \quad (56)$$

$$\sigma(R_1, R_2) = \beta_0 + \beta_1 R_1 + \beta_2 \sqrt{R_2}, \quad (57)$$

$$R_{n,t} := \int_{-\infty}^t K_{\alpha_n, \delta_n}(t-u) \left(\frac{dS_u}{S_u} \right)^n, \quad K_{\alpha_n, \delta_n}(\tau) = \frac{1}{Z_{\alpha_n, \delta_n}} (\tau + \delta_n)^{-\alpha_n}, \quad (58)$$

for $n \in \{1, 2\}$, with $Z_{\alpha_n, \delta_n} = \delta_n^{1-\alpha_n} / (\alpha_n - 1)$.

The TSPL kernel admits Laplace transform

$$\frac{1}{x^\alpha} = \frac{1}{\Gamma(\alpha)} \int_0^\infty \lambda^{\alpha-1} e^{-\lambda x} d\lambda, \quad x > 0, \alpha > 0. \quad (59)$$

Substitute $x = \tau + \delta$:

$$(\tau + \delta)^{-\alpha} = \frac{1}{\Gamma(\alpha)} \int_0^\infty \lambda^{\alpha-1} e^{-\lambda \delta} e^{-\lambda \tau} d\lambda.$$

Divide by $Z_{\alpha,\delta} = \delta^{1-\alpha}/(\alpha-1)$ and use $\Gamma(\alpha) = (\alpha-1)\Gamma(\alpha-1)$:

$$K_{\alpha,\delta}(\tau) = \int_0^\infty \lambda e^{-\lambda\tau} \cdot \frac{\delta^{\alpha-1}}{\Gamma(\alpha-1)} \lambda^{\alpha-2} e^{-\lambda\delta} d\lambda. \quad (60)$$

Substitute the mixture into the definition of $R_{n,t}$:

$$R_{n,t} = \int_{-\infty}^t \left[\int_0^\infty \lambda e^{-\lambda(t-u)} \cdot \frac{\delta_n^{\alpha_n-1}}{\Gamma(\alpha_n-1)} \lambda^{\alpha_n-2} e^{-\lambda\delta_n} d\lambda \right] \left(\frac{dS_u}{S_u} \right)^n.$$

By applying Fubini's theorem we get

$$R_{n,t} = \int_0^\infty F_{n,t}^\lambda \cdot \frac{\delta_n^{\alpha_n-1}}{\Gamma(\alpha_n-1)} \lambda^{\alpha_n-2} e^{-\lambda\delta_n} d\lambda, \quad F_{n,t}^\lambda := \int_{-\infty}^t \lambda e^{-\lambda(t-u)} \left(\frac{dS_u}{S_u} \right)^n. \quad (61)$$

Write $F_{n,t}^\lambda = \lambda e^{-\lambda t} \int_{-\infty}^t e^{\lambda u} (dS_u/S_u)^n$ and use Ito's product rule

$$dF_{n,t}^\lambda = \lambda \left[\left(\frac{dS_t}{S_t} \right)^n - F_{n,t}^\lambda dt \right].$$

Substituting $(dS_t/S_t)^1 = \sigma_t dW_t$ and $(dS_t/S_t)^2 = \sigma_t^2 dt$:

$$dF_{1,t}^\lambda = \lambda(\sigma_t dW_t - F_{1,t}^\lambda dt), \quad dF_{2,t}^\lambda = \lambda(\sigma_t^2 - F_{2,t}^\lambda) dt. \quad (62)$$

Each $F_{n,t}^\lambda$ is a stationary Ornstein–Uhlenbeck process with mean-reversion rate λ . Representing the TSPL kernel as a continuum of such OU factors is inspired by Muravlev^[11].

Now continue with $d\sigma_t$ in three steps: $dR_{1,t}$, $dR_{2,t}$, then differentiate $\sigma_t = \beta_0 + \beta_1 R_{1,t} + \beta_2 \sqrt{R_{2,t}}$.

Step 1. Dynamics of $R_{1,t}$.

The only changing part is $F_{1,t}^\lambda$ with $dF_{1,t}^\lambda = \lambda(\sigma_t dW_t - F_{1,t}^\lambda dt)$. This yields

$$dR_{1,t} = \int_0^\infty \lambda(\sigma_t dW_t - F_{1,t}^\lambda dt) \cdot \frac{\delta_1^{\alpha_1-1}}{\Gamma(\alpha_1-1)} \lambda^{\alpha_1-2} e^{-\lambda\delta_1} d\lambda.$$

Observe that

$$\int_0^\infty \lambda \cdot \frac{\delta_1^{\alpha_1-1}}{\Gamma(\alpha_1-1)} \lambda^{\alpha_1-2} e^{-\lambda\delta_1} d\lambda = \frac{\alpha_1-1}{\delta_1}.$$

Further define

$$\bar{\lambda}_1 := \frac{\alpha_1-1}{\delta_1}, \quad \bar{R}_{1,t} := \int_0^\infty F_{1,t}^\lambda \cdot \frac{\delta_1^{\alpha_1-1}}{\Gamma(\alpha_1-1)} \lambda^{\alpha_1-2} e^{-\lambda\delta_1} d\lambda. \quad (63)$$

Pulling $\sigma_t dW_t$ out of the λ -integral and using the introduced notation

$$dR_{1,t} = \bar{\lambda}_1(\sigma_t dW_t - \bar{R}_{1,t} dt). \quad (64)$$

Step 2. Dynamics of $R_{2,t}$.

The same procedure with $dF_{2,t}^\lambda = \lambda(\sigma_t^2 - F_{2,t}^\lambda) dt$

$$dR_{2,t} = \int_0^\infty \lambda(\sigma_t^2 - F_{2,t}^\lambda) dt \cdot \frac{\delta_2^{\alpha_2-1}}{\Gamma(\alpha_2-1)} \lambda^{\alpha_2-2} e^{-\lambda\delta_2} d\lambda.$$

Introduce

$$\bar{\lambda}_2 := \frac{\alpha_2 - 1}{\delta_2}, \quad \bar{R}_{2,t} := \int_0^\infty F_{2,t}^\lambda \cdot \frac{\delta_2^{\alpha_2}}{\Gamma(\alpha_2)} \lambda^{\alpha_2-1} e^{-\lambda \delta_2} d\lambda, \quad (65)$$

and therefore

$$dR_{2,t} = \bar{\lambda}_2(\sigma_t^2 - \bar{R}_{2,t}) dt. \quad (66)$$

Step 3. *Derive $d\sigma_t$.*

$R_{2,t}$ is finite-variation, so the chain rule gives

$$d\sqrt{R_{2,t}} = \frac{dR_{2,t}}{2\sqrt{R_{2,t}}} = \frac{\bar{\lambda}_2(\sigma_t^2 - \bar{R}_{2,t})}{2\sqrt{R_{2,t}}} dt.$$

Differentiating $\sigma_t = \beta_0 + \beta_1 R_{1,t} + \beta_2 \sqrt{R_{2,t}}$ and substituting the two previous steps,

$$d\sigma_t = \left(-\beta_1 \bar{\lambda}_1 \bar{R}_{1,t} + \frac{\beta_2 \bar{\lambda}_2}{2} \frac{\sigma_t^2 - \bar{R}_{2,t}}{\sqrt{R_{2,t}}} \right) dt + \beta_1 \bar{\lambda}_1 \sigma_t dW_t. \quad (67)$$

This is structurally identical to the $d\sigma_t$ SDE (54).

Remark 5.10.

1. We recover Guyon's n -EXP SDE form, just with infinitely many OU factors. The n -EXP truncation is in dimension, not structure.
2. The lognormal vol-of-vol coefficient is explicit and has meaningful dynamic interpretation. δ_1 sets the attention horizon. Large δ_1 smoothens attention through all past returns and damps vol-of-vol.

We finalize this subsection by providing fully Markovian dynamics.

The Markovian system.

Initializing in the state of $(S_t, F_{1,t}^{\lambda \in (0,\infty)}, F_{2,t}^{\lambda \in (0,\infty)})$ one can further simulate it by using

$$\frac{dS_t}{S_t} = \sigma_t dW_t \quad (68)$$

$$d\sigma_t = \left(-\beta_1 \bar{\lambda}_1 \bar{R}_{1,t} + \frac{\beta_2 \bar{\lambda}_2}{2} \frac{\sigma_t^2 - \bar{R}_{2,t}}{\sqrt{R_{2,t}}} \right) dt + \beta_1 \bar{\lambda}_1 \sigma_t dW_t. \quad (69)$$

where computation of σ_t , $R_{1,t}$, $R_{2,t}$ and $\bar{R}_{1,t}$, $\bar{R}_{2,t}$ comes from

$$\sigma_t = \beta_0 + \beta_1 R_{1,t} + \beta_2 \sqrt{R_{2,t}}, \quad (70)$$

$$R_{n,t} = \bar{\lambda}_n \int_0^\infty \frac{F_{n,t}^\lambda}{\lambda} w_n(\lambda) d\lambda, \quad F_{n,t}^\lambda := \int_{-\infty}^t \lambda e^{-\lambda(t-u)} \left(\frac{dS_u}{S_u} \right)^n. \quad (71)$$

$$\bar{R}_{1,t} = \int_0^\infty F_{1,t}^\lambda \cdot w_1(\lambda) d\lambda, \quad w_1(\lambda) = \frac{\delta_1^{\alpha_1}}{\Gamma(\alpha_1)} \lambda^{\alpha_1-1} e^{-\lambda \delta_1} \quad (72)$$

$$\bar{R}_{2,t} = \int_0^\infty F_{2,t}^\lambda \cdot w_2(\lambda) d\lambda, \quad w_2(\lambda) = \frac{\delta_2^{\alpha_2}}{\Gamma(\alpha_2)} \lambda^{\alpha_2-1} e^{-\lambda \delta_2} \quad (73)$$

the update of the system comes from

$$dF_{1,t}^\lambda = \lambda(\sigma_t dW_t - F_{1,t}^\lambda dt), \quad \lambda \in (0, \infty) \quad (74)$$

$$dF_{2,t}^\lambda = \lambda(\sigma_t^2 - F_{2,t}^\lambda) dt, \quad \lambda \in (0, \infty). \quad (75)$$

Note the system is Markovian in $(S_t, F_{1,t}^{\lambda \in (0,\infty)}, F_{2,t}^{\lambda \in (0,\infty)})$.

6 Fitting

Before going to forecasts, it is reasonable to understand where the model lacks in its fitting procedure. The purpose of this section is to identify shortcomings of Guyon’s PDV model in fit, address them and proceed accordingly.

6.1 Motivation for forecasting

6.1.1 Fit vs forecast

The question that Guyon’s model answers is “Given today’s closing price of SPX and past dynamics, how much of today’s closing VIX value can I explain?”. In this sense, the model is a fitting tool, not a forecasting one. It explains the VIX in mean, but says nothing about the dynamics that a trader needs. The model is also reliant on the latest-day return: without it the dynamics break.

Guyon and Lekeufack^[7] report the fitted parameters

$$\beta_0 = 0.057, \quad \beta_1 = -0.095, \quad \beta_2 = 0.82, \quad \alpha_1 = 1.06, \quad \delta_1 = 0.020, \quad \alpha_2 = 1.60, \quad \delta_2 = 0.052. \quad (76)$$

By computing the empirical truncated TSPL kernel, the weights of the most recent returns for $R_{1,t}$ and $R_{2,t}$ become:

$$W_1 = 4.28\%, \quad W_2 = 4.82\%. \quad (77)$$

The reliance on the most recent return is confirmed by the following question:

“If at the beginning of day t one wants to forecast the VIX close at day t , without seeing the open-close return for day t , how much quality and dynamics degrade?” The graphs below answer it.

Method A

The *morning prediction* sets $r_t = 0$ in the kernel anchored at t . The morning prediction becomes a shifted copy of the same-day fit: without the new return the model leans on the last available one. Both quality and dynamics degrade. Provides no valuable forecasting information.

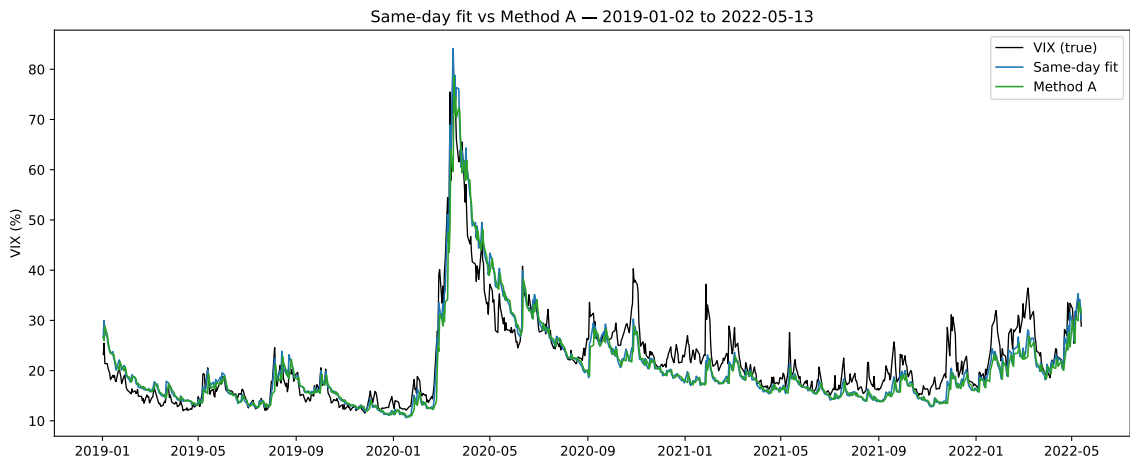


Figure 1: Realised VIX, same-day fit, and morning prediction on the 2019–2022 test window. Same-day RMSE = 3.43%; morning-prediction RMSE = 4.20% ($1.22\times$ worse).

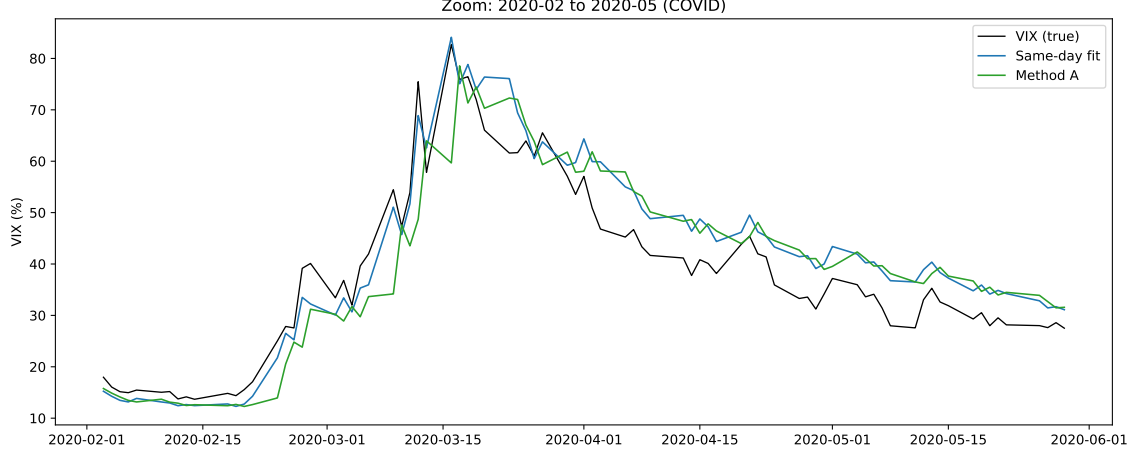


Figure 2: COVID zoom: 2020-02 – 2020-05.

This is the first justification to reframe the model if one demands forecasting quality.

6.1.2 Trajectory of information matters

Guyon’s original paper is named “Volatility is (mostly) path-dependent”^[7], suggesting that explanation comes from incorporating past information. We propose a stronger statement: the trajectory of the returns matters, not merely the daily endpoint they reach. The same daily return assembled from different intraday paths produces a different σ_{t+1} .

Numerical example

1. Take Guyon’s 4-factor 2EXP parametrisation: $\lambda_{1,j} = (55, 10)$, $\lambda_{2,j} = (20, 3)$, $\theta_1 = 0.25$, $\theta_2 = 0.5$, $\beta_0 = 0.04$, $\beta_1 = -0.13$, $\beta_2 = 0.65$.
2. Initialise $R_{n,j,0}$ from real SPX history at 2010-05-01 via the integral form of (49), truncated at the past 1000 trading days; this yields $\sigma_0 = 21.24\%$.
3. Fix the daily return $r_t = -5\%$.
4. Compute $d\sigma_t$ two ways: **(a)** one daily step over $[t-1, t]$; **(b)** three intraday substeps with sub-returns -2% , $+3\%$, -5.88% , so that the daily return remains -5% .
5. Observe the difference.

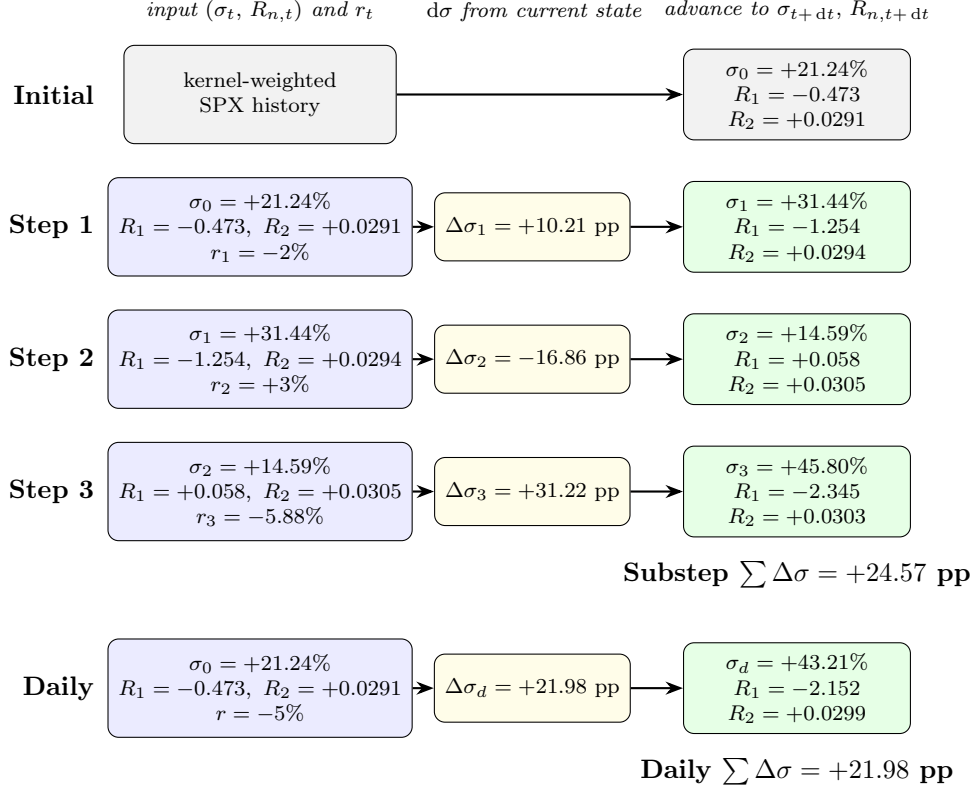


Figure 3: Per-substep evolution of σ via Euler discretisation of Guyon's 4-factor 2EXP SDE, eqs. (46)–(54). Both paths share the same daily return (-5%). Difference is equal to 2.59 pp. Real market index change is quoted in 5 pp.

6.1.3 Information delay

A textbook definition of the efficient-markets hypothesis is that the price process is a martingale on the filtration $\mathcal{F}_t$:

$$\mathbb{E}^{\mathbb{Q}}[S_{t+u} | \mathcal{F}_t] = S_t, \quad u > 0. \quad (78)$$

The expected return is zero on every horizon, and any forecasting attempt is reduced to noise. We propose a weaker statement. There exists $\varepsilon > 0$ such that

$$\mathbb{E}^{\mathbb{Q}}[S_{t+u} | \mathcal{F}_{t-\varepsilon}] = S_t, \quad u > 0. \quad (79)$$

Equation (79) is a martingale on the lagged filtration $\mathcal{F}_{t-\varepsilon}$. At time t the market has absorbed information only up to $t - \varepsilon$, so S_t is $\mathcal{F}_{t-\varepsilon}$ -measurable. A trader fast enough to observe $\mathcal{F}_t$ sees the ε -window $(t - \varepsilon, t]$ of news the price has not yet incorporated; extracting signal from this window yields a statistically positive edge.

Defining ε is a completely different beast, but disregarding intraday information should give a worse view of the market than holding it.

6.2 Error analysis

To build a predicting model, one must first understand where the fitting model is mistaken. We start with an error analysis, which suggests what the model is missing and how it can

be fixed.

We refit Guyon’s extended SPX/VIX data (train 2000-01-01 to 2021-12-31, test 2022-01-01 to 2026-01-01) and inspect the residual $\varepsilon_t = \text{VIX}_t^{\text{actual}} - \text{VIX}_t^{\text{fit}}$ along with its rolling z -score $z_t = (\varepsilon_t - \mu_t^{252})/\sigma_t^{252}$, where μ_t^{252} and σ_t^{252} are the trailing-252-day mean and standard deviation of ε .

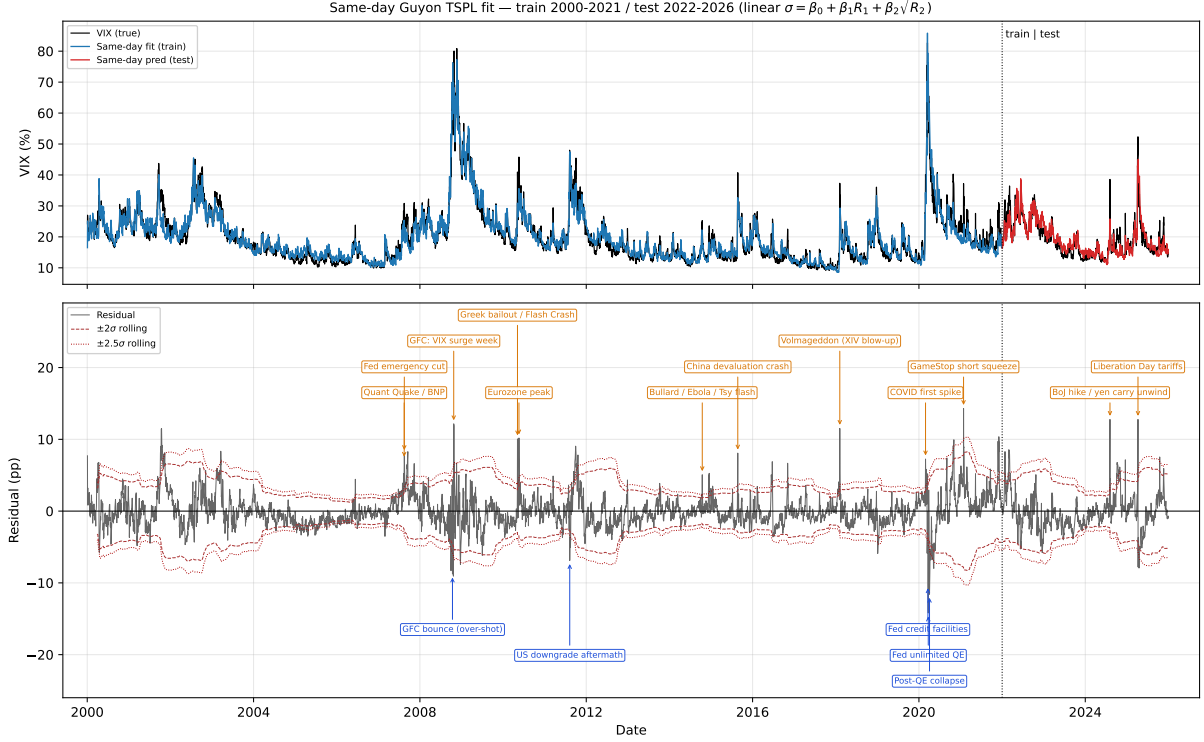


Figure 4: Same-day Guyon TSPL fit on extended data. Residuals plot.

17 events catalogued in Table 1.

Table 1: Seventeen macro events flagged on the residual panel of Figure 4

#	Date	Event	Description	Why model missed
1	2007-08-16	Quant Quake / BNP	BNP froze 3 funds; Quant strategies forced unwind.	Bank-fund freeze, exogenous
2	2007-08-17	Fed emergency cut	Fed cut discount rate 50 bp to signal help. Markets regarded it as a problem.	Exogenous Fed announcement
3	2008-10-13	GFC drop	Worst week ended; +11% day; VIX stayed elevated despite positive returns.	Market uncertainty blunts features
4	2008-10-24	GFC: VIX surge	TARP stuck; fresh credit-market freeze; VIX surged past prior peak.	R_2 saturated, credit-driven
5	2010-05-07	Greek bailout / Flash Crash	Day after Flash Crash; Greek bailout doubts; VIX stayed at panic levels.	Foreign shock
6	2010-05-20	Eurozone crisis	Germany unilateral short-sale ban; Eurozone bank stocks plunged; VIX surged.	Foreign shock
7	2011-08-09	US downgrade aftermath	S&P downgraded US Aug 5; Spike, then rapid Fed-soothed collapse.	FOMC dovish surprise
8	2014-10-16	Bullard / Ebola / Tsy flash	Bullard QE pivot, Ebola panic, Treasury 30 bp intraday flash crash.	Triple exogenous shock
9	2015-08-24	China devaluation crash	China devalued yuan; US opened limit-down $\rightarrow$ VIX hit 50.	China FX policy chain
10	2018-02-05	Volmageddon	XIV/SVXY rebalance feedback loop; VIX from 17 to 37 in one day.	Market feedback in VIX futures
11	2020-02-28	COVID first spike	Coronavirus spread to US confirmed; VIX jumped to 40.	Pandemic news
12	2020-03-20	Fed credit facilities	Fed announced monetary help; Vol started compressing.	Fed policy outpaces R_2
13	2020-03-23	Fed unlimited QE	Fed announced unlimited Treasury/MBS purchases	Fed bazooka, R_2 saturated
14	2020-04-03	Post-QE collapse	Continued vol compression post-Fed	Slow R_2 decay over-projects
15	2021-01-27	GameStop short squeeze	Gamestop short squeeze; Hedge-fund liquidation fears; VIX from 23 to 37.	Single-stock retail squeeze
16	2024-08-05	BoJ hike, Yen carry unwind	BoJ raised rates Jul 31; Yen carry-trade unwind force-sold US equities; VIX hit 65.	Cross-asset carry-trade unwind
17	2025-04-08	Liberation Day tariffs	Trump announced tariffs; VIX peaked Apr 8.	Tariff policy announcement

The 17 residuals admit two complementary views: by shock type (Figure 5) and by what would resolve each (Figure 6).

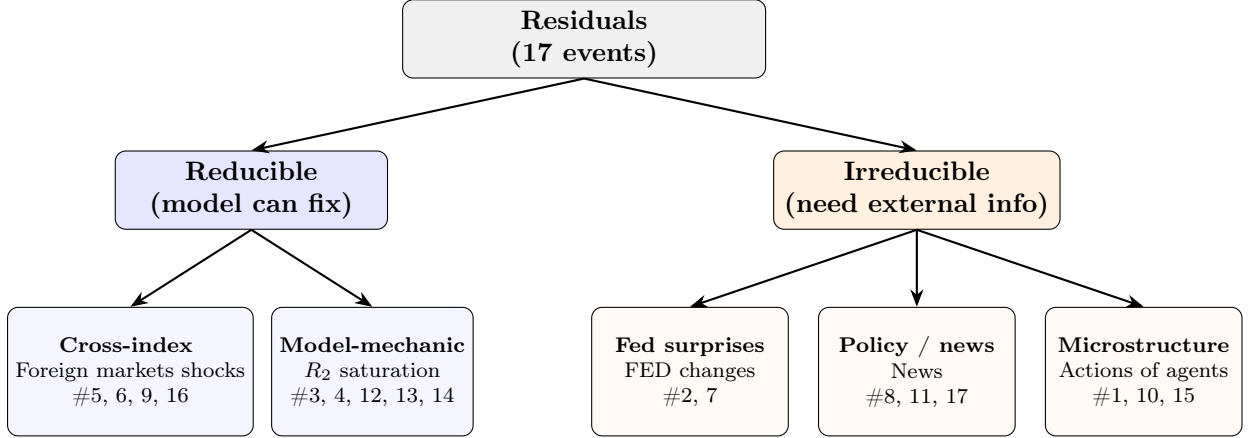


Figure 5: Residual by shock type. Reducible categories admit a model-side fix; irreducible categories require information outside any market-priced feature.

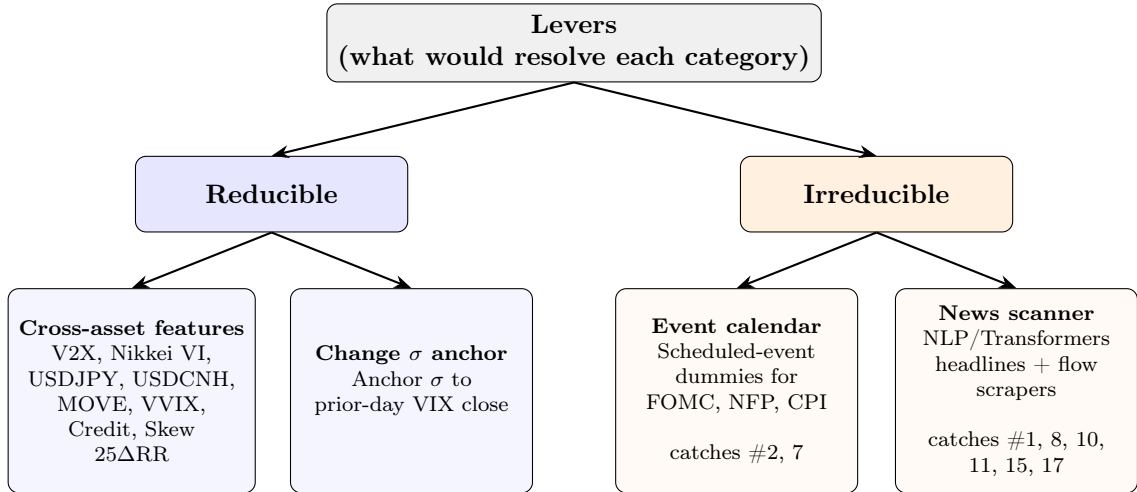


Figure 6: Levers that would resolve each residual category.

Candidate features to include:

Summary.

This subsection analyses the sources of error and potential improvements. The two cheapest leverage points are:

1. Adding VIX_{t-1} ensures non-saturation of $R_{2,t}$. Limits potential static error, while dynamics of the model remains meaningful
2. Adding intraday and overnight dynamics. Toy example shows why it is meaningful.

6.3 Empirical model comparison

The two cheapest leverage points are the previous VIX level and intraday dynamics. We test both on a fixed window (2023-07-19 to 2025-09-16); calibration uses extended training

Table 2: Proposed feature classes addressing the reducible residual categories of Figure 5.

Feature	What it is	How computed	How it helps
VVIX	Vol of VIX	30-day IV of VIX options	Vol-of-vol regime signal
VIX term slope	Ratio across vol maturities	VIX9D/VIX or VX1/VX2 futures	Backwardation flags stress
SKEW	Tail-risk premium	RN-density 3rd moment from OTM SPX options	OTM-put premium leads spike
25 Δ RR	Put-call IV asymmetry	IV(25 Δ call) – IV(25 Δ put)	Hedger flow leads spot
MOVE	Treasury implied vol	1M IV of UST 2/5/10/30y options	Rates stress leads equities
USDJPY signals	Yen carry indicators	Spot, 1M RR, 3M xccy basis	Carry unwind leads SPX
Credit spreads	Corporate default premium	HY OAS, CDX HY, HYG–LQD	Default risk leads vol
Foreign vol indices	EU & Asia implied vol	V2X (Eurostoxx), VNKY (Nikkei)	Foreign session leads NY
Foreign index returns	EU & Asia equity returns	TSPL kernel on SX5E, NKY returns	Extends PDV core; improves fit
Event dummies	Macro-calendar flags	Binary on FOMC, NFP, CPI dates	Catches scheduled Fed days

data 1990-01-01 to 2023-07-18. We also add European and Japanese index returns to roughly assess the effect of other markets.

6.3.1 Models

1. Models 1, 3, 4, 5 share the Guyon 4-factor 2EXP form $\sigma_t = \beta_0 + \beta_1 R_{1,t} + \beta_2 \sqrt{R_{2,t}}$ with $R_{n,j,t}$ updated via the Markov scheme of (46-54) on intraday scale.
2. Model 2 is a simple random walk.
3. Model 5 freezes $(\beta_0, \beta_1, \beta_2, \lambda_{n,j}, \theta_n)$ from model 1 and OLS-fits $\beta_3, \dots, \beta_6$. Reusing SPX (λ, θ) for SX5E and NKY assumes the kernel timescales transfer across equity indices. It is a rough approximation to deal with potential dimensionality issues that still yields results. Guyon^[7] reports that (λ, θ) are surprisingly similar across different US indexes.

Foreign returns are local-currency close-to-close on the SPX trading calendar. NKY closes 01–02 ET and SX5E closes 10:30–11:30 ET, both before the SPX 16:00 ET close, so potential leakage is absent.

6.3.2 Figures

6.3.3 Summary

1. All leverage points from §6.2 yield results.
2. Daily multi-index features (m5) cut RMSE from 2.61% to 2.34%, lift R^2 from 0.687 to 0.749 and direction-correct from 0.55 to 0.61 over SPX-only.

Table 3: (a) Five models

#	Name	Mechanism	Information used
1	Guyon SPX-only	$VIX_t^{\text{close}} = \beta_0 + \beta_1 R_1^{\text{SPX}} + \beta_2 \sqrt{R_2^{\text{SPX}}}$	Daily SPX returns through close
2	VIX random walk	$VIX_t^{\text{close}} = VIX_{t-1}^{\text{close}}$	Yesterday's VIX close
3	Today VIX_{open} + intraday SDE	$VIX_t^{\text{close}} = VIX_t^{\text{open}} + \sum_{t_{\text{open}}}^{t_{\text{close}}} \Delta VIX$	Today $VIX_{10:00}$ + 6 hourly SPX returns
4	$VIX_{\text{close}}^{\text{yesterday}}$ + overnight + intraday SDE	$VIX_t^{\text{close}} = VIX_{t-1}^{\text{close}} + \sum_{t-1}^{t_{\text{close}}} \Delta VIX$	Yesterday VIX_{close} + overnight + 6 hourly SPX
5	Multi-index Guyon	$VIX_t^{\text{close}} = \beta_0 + \beta_1 R_1^{\text{SPX}} + \beta_2 \sqrt{R_2^{\text{SPX}}} + \beta_3 R_1^{\text{SX5E}} + \beta_4 \sqrt{R_2^{\text{SX5E}}} + \beta_5 R_1^{\text{NKY}} + \beta_6 \sqrt{R_2^{\text{NKY}}}$	Daily SPX + SX5E + NKY. SPX kernel (λ, θ) reused for foreign factors.

Figure 7: (b) Each panel compares the m1 baseline (grey) with one model (colour) against actual VIX_{close} (black).

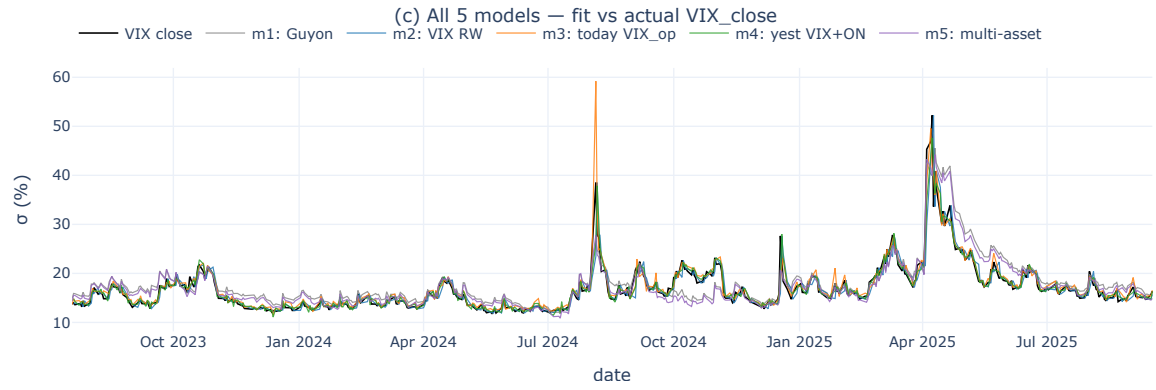


Figure 8: (c) All five fits against actual VIX_{close} .

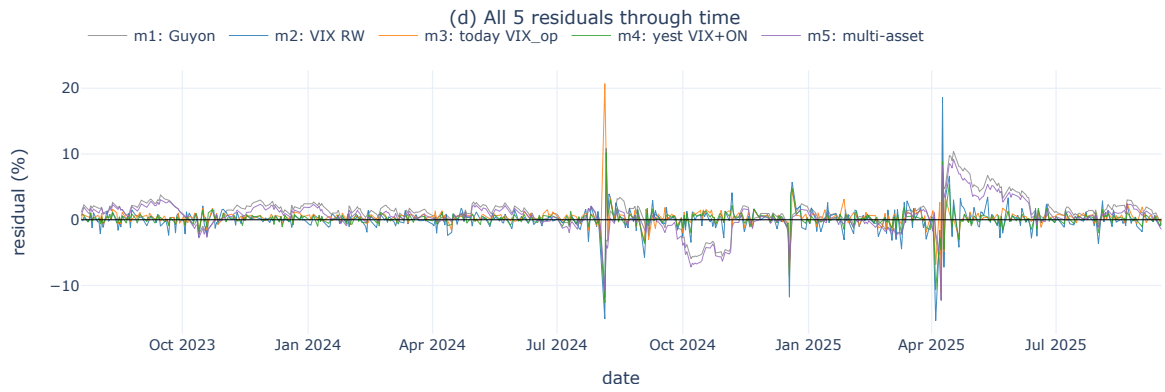


Figure 9: (d) Residuals $\hat{VIX}_t - VIX_t^{close}$ through time.

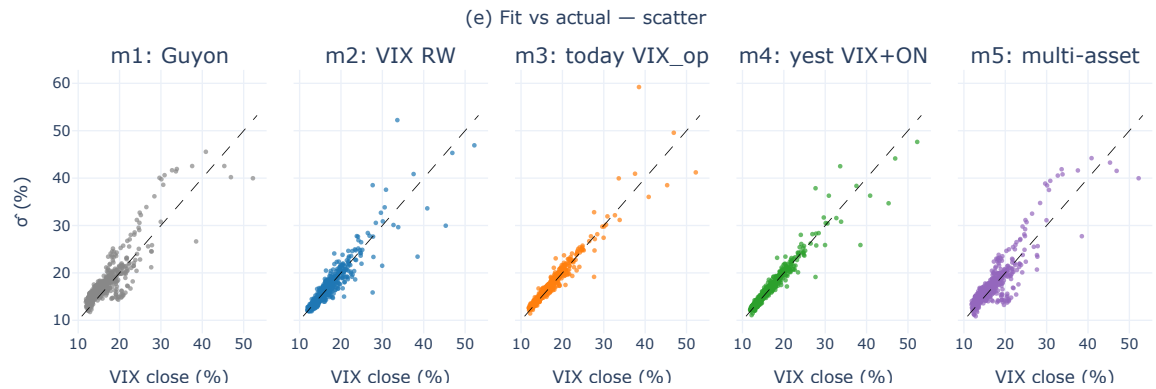


Figure 10: (e) Fit vs actual scatter

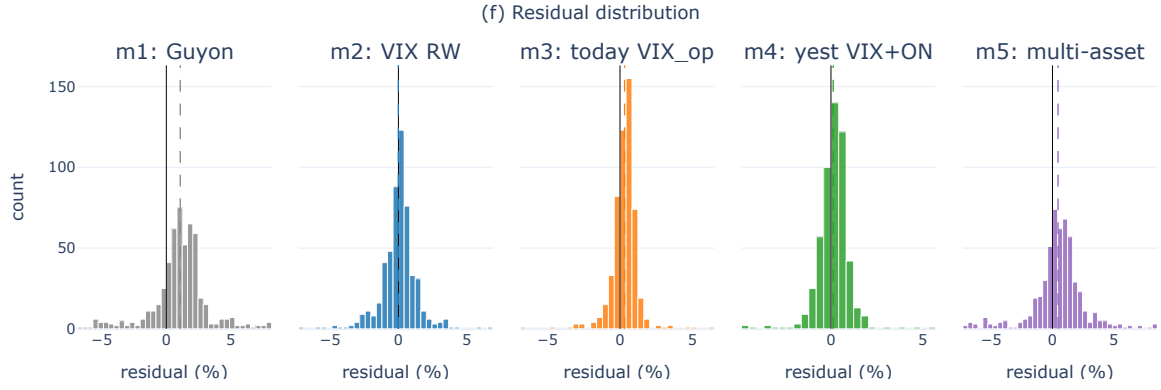


Figure 11: (f) Residual distribution per model.

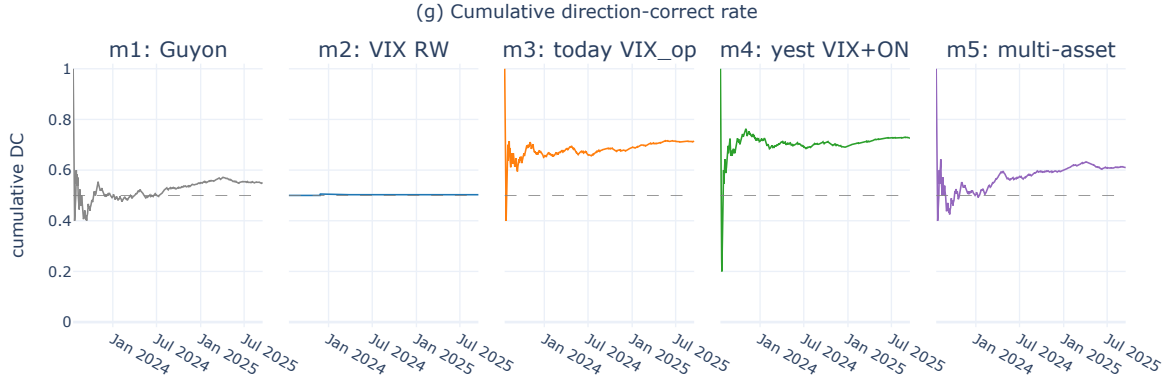


Figure 12: (g) Cumulative direction-correct rate, $DC_t = \frac{1}{N_t} \sum_{s \leq t} \mathbf{1}\{\text{sign}(\hat{VIX}_t - VIX_{s-1}^{\text{close}}) = \text{sign}(VIX_s^{\text{close}} - VIX_{s-1}^{\text{close}})\}$. Model 2 always yields $\text{sign}=0$. Counted as 0.5

Table 4: (h) Test-window summary metrics. 538 sessions in total; $n = 537$ for models that require a previous-day VIX anchor.

	Model	RMSE	MAE	bias	R^2	DC	n
m1	Guyon SPX-only	2.610%	1.908%	+1.076%	0.687	0.550	538
m2	VIX random walk	1.957%	1.016%	-0.005%	0.824	0.503	537
m3	Today VIX_{op} + intraday	1.459%	0.749%	+0.315%	0.902	0.714	538
m4	Yest VIX + ON + intraday	1.323%	0.695%	+0.121%	0.920	0.727	537
m5	Multi-asset Guyon	2.336%	1.564%	+0.454%	0.749	0.610	538

3. Pure intraday SDE with VIX anchored to open drops RMSE to 1.46%, lifts R^2 above 0.90, and direction-correct from 0.55 to 0.71.
4. Adding overnight dynamics and anchoring VIX to the previous close drops RMSE to 1.32%, lifts R^2 to 0.92, and direction-correct to 0.73.
5. The random walk explains 82% of the variance, yet says nothing about dynamics. For option pricing it would fail on dynamic error.
6. The lift from m1 to m5 suggests the foreign block carries information m4 does not have.

7 Forecasting. SDE Approach

The previous section fits daily-close VIX from the path-dependent state on the test window. This section tests whether the same model transports to intraday forecasting of VX futures and produces a profitable strategy under realistic execution.



Figure 13: ES and VX mid-price over 2025-03-02 – 2025-08-29

7.1 Data used

The backtest draws on two data sources. FirstRateData (FRD) supplies daily OHLCV on cash SPX and front-month VX futures for offline calibration. Databento supplies MBP-10 limit-order-book ticks on ES and VX futures for the intraday backtest.

Table 5: Order-book data sources and instrument characteristics for the two products studied in this section.

	ES	VX
Continuous symbol	ES.c.0	VX.c.0
Exchange / venue	CME Globex	Cboe Futures Exchange (CFE)
Underlying index	S&P 500	VIX
Notional multiplier	$\$50 \times \text{SPX}$	$\$1,000 \times \text{VIX}$
Tick size	0.25 index pt (\$12.50)	0.05 vol pt (\$50)
Approx. notional at quote	\$300k (SPX $\sim$ 6,000)	\$20k (VIX $\sim$ 20)
Main session (RTH)	Mon–Fri 09:30 – 16:15 ET	Mon–Fri 09:30 – 16:00 ET
Extended session (ETH)	Sun 18:00 – Fri 17:00 ET	Sun 18:00 – Fri 16:00 ET
Vendor	Databento	Databento
Dataset	GLBX.MDP3	XCBF.PITCH
Schema	mbp-10	mbp-10
Data window	2025-03-02 – 2025-08-29	2025-03-02 – 2025-08-29
Contracts active in window	ESH5, ESM5, ESU5	VXH5, VXJ5, VXX5, VXM5, VZN5, VXQ5, VXU5

Table 6: Cash-settlement schedule for contracts present in the data window. Last-trade is the moment the expiring contract stops trading and cash settlement is computed. *Underlying ref. (ET)* is the timestamp at which the index value used in the SOQ calculation is itself observed (exactly 30 calendar days after the VX expiry Wednesday).

Contract	Last-trade (ET)	Underlying ref. (ET)	In window?
ESH5	Fri 2025-03-21, 09:30	Fri 2025-03-21, 09:30	yes
ESM5	Fri 2025-06-20, 09:30	Fri 2025-06-20, 09:30	yes
ESU5	Fri 2025-09-19, 09:30	Fri 2025-09-19, 09:30	no
VXH5	Wed 2025-03-19, 09:00	Fri 2025-04-18, 09:30	yes
VXJ5	Wed 2025-04-16, 09:00	Fri 2025-05-16, 09:30	yes
VXK5	Wed 2025-05-21, 09:00	Fri 2025-06-20, 09:30	yes
VXM5	Wed 2025-06-18, 09:00	Fri 2025-07-18, 09:30	yes
VXN5	Wed 2025-07-16, 09:00	Fri 2025-08-15, 09:30	yes
VXQ5	Wed 2025-08-20, 09:00	Fri 2025-09-19, 09:30	yes
VXU5	Wed 2025-09-17, 09:00	Fri 2025-10-17, 09:30	no

FRD data spans 2008-07-10 through 2025-02-28; the first four years serve as the kernel’s lookback, and Guyon’s model is fit on the full 2012-07-10–2025-02-28 window. Fitting on VX rather than VIX calibrates directly to the traded instrument; the basis is tight, with mean absolute basis 1.53 vol points and Pearson correlation 0.97 across 3,177 daily bars.

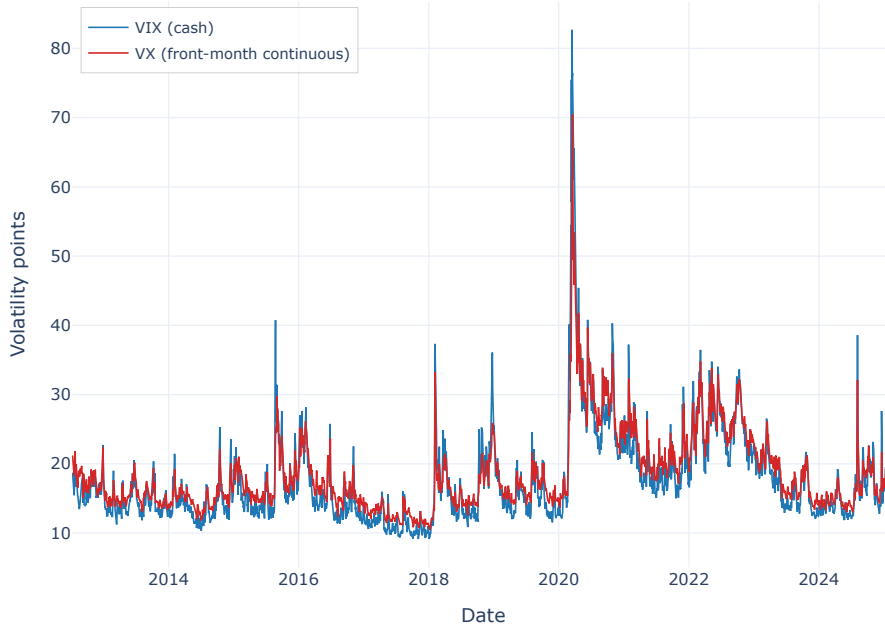


Figure 14: Cash VIX (blue) and FirstRateData front-month continuous VX (red) over the calibration period 2012-07-10 – 2025-02-28 on a shared volatility-points axis.

Databento covers 2025-03-02 through 2025-08-29. ES stands in for the SPX-return driver dS_t/S_t that propagates the PDV state since cash SPX is unavailable at minute resolution from Databento. ES tracks SPX with mean absolute basis 8.20 index points and Pearson correlation 0.9999 over the FRD calibration window (Figure 15).

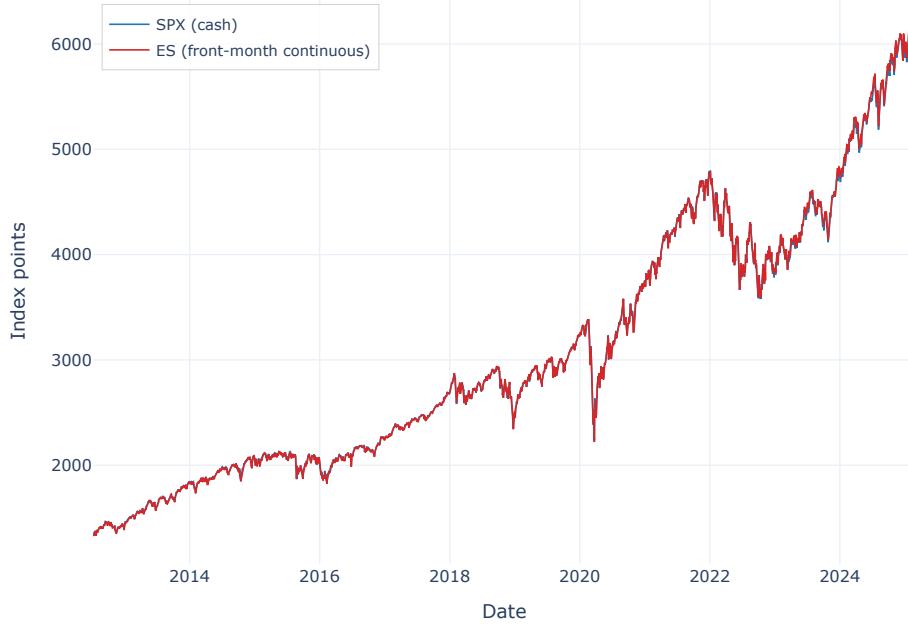


Figure 15: Cash SPX (blue, FirstRateData) and front-month continuous ES (red, yfinance) over the calibration period 2012-07-10 – 2025-02-28.

7.2 Model used

We use the four-factor Markovian PDV model, propagated at one-minute resolution during the RTH and ETH. The defining equations are restated below; all symbols carry the meaning given in §5.

$$\frac{dS_t}{S_t} = r_t. \quad (80)$$

$$R_{1,t} = (1 - \theta_1) R_{1,0,t} + \theta_1 R_{1,1,t}. \quad (81)$$

$$R_{2,t} = (1 - \theta_2) R_{2,0,t} + \theta_2 R_{2,1,t}. \quad (82)$$

$$dR_{1,j,t} = \lambda_{1,j} (r_t - R_{1,j,t}) dt, \quad j \in \{0, 1\}. \quad (83)$$

$$dR_{2,j,t} = \lambda_{2,j} (r_t^2 - R_{2,j,t}) dt, \quad j \in \{0, 1\}. \quad (84)$$

$$d\sigma_t = \left(-\beta_1 \bar{\lambda}_1 \bar{R}_{1,t} + \frac{\beta_2 \bar{\lambda}_2}{2} \frac{\sigma_t^2 - \bar{R}_{2,t}}{\sqrt{R_{2,t}}} \right) dt + \beta_1 \bar{\lambda}_1 \sigma_t dW_t. \quad (85)$$

The directional forecast at time t over horizon H is the conditional expectation $\mathbb{E}[d\sigma_t | \mathcal{F}_t] = \mu_t dt$.

Remark 7.1. The driver dS_t/S_t in the state update is the ES one-minute return.

7.3 Calibration and Propagation

Calibration

The parameter vector $\Theta = (\lambda_{1,0}, \lambda_{1,1}, \lambda_{2,0}, \lambda_{2,1}, \beta_0, \beta_1, \beta_2, \theta_1, \theta_2)$ is fit once on VX and SPX (FirstRateData) over 2012-07-10 – 2025-02-28 by Guyon & Lekeufack’s^[7] two-step procedure. Table 7 reports $\hat{\Theta}$ and the training R^2 .

Table 7: Fitted $\hat{\Theta}$ for the backtest. Train $R^2 = 0.893$.

$\lambda_{1,0}$	16.98	$\lambda_{1,1}$	13.50	$\lambda_{2,0}$	12.14
$\lambda_{2,1}$	1.36	β_0	0.0502	β_1	−0.0451
β_2	0.755	θ_1	0.0131	θ_2	0.779

Propagator

At the 2025-02-28 close the propagator state is initialized by kernel-summing equations (83)–(84) over the last 1000 daily FRD SPX returns. The model then switches to VX/ES data at one-minute resolution, taking the mid-price during both RTH and ETH.

Table 8: State trajectory from initialization to the first boundary.

timestamp	Δt (min)	ES mid	VX mid	$\bar{R}_1$	$\bar{R}_2$	$\sqrt{\bar{R}_2}$
2025-02-28 16:00 ET (init)	–	5954.50	18.83	−0.1853	0.0193	0.1390
2025-03-02 08:00 ET (bridge)	2400	5958.13	18.83	−0.1564	0.0202	0.1420
2025-03-02 18:00 ET	1	5970.75	18.68	−0.1161	0.0204	0.1428
2025-03-02 23:59 ET	1	5971.88	18.73	−0.1107	0.0205	0.1432
2025-03-03 09:30 ET (RTH)	1	5982.38	18.53	−0.0785	0.0207	0.1439
2025-03-03 10:30 ET	1	5965.38	18.78	−0.1265	0.0207	0.1439

7.4 Signal generation

Trading runs only from 10:30 to 15:00 ET to exclude the U-shape of the intraday volume profile.

At the beginning of each window, the signal is generated as the expectation of equation (85):

$$\mathbb{E}[\mathrm{d}\sigma_t \mid \mathcal{F}_t] = \left(-\beta_1 \bar{\lambda}_1 \bar{R}_{1,t} + \frac{\beta_2 \bar{\lambda}_2}{2} \frac{\sigma_t^2 - \bar{R}_{2,t}}{\sqrt{\bar{R}_{2,t}}} \right) \mathrm{d}t = \mu_t \mathrm{d}t. \quad (86)$$

The H -ahead forecast in VX points is

$$f_t(H) = 100 \cdot \mu_t \cdot \frac{H}{252 \cdot 1440}, \quad (87)$$

we evaluate $H \in \{1, 5, 10, 15, 30, 60\}$ minutes and gate by top 5, 10 or 20% of magnitude.

7.5 Position execution

1. Assume $t = 10:30$ ET and the signal stated long. Best bid = 45.60, best ask = 45.65, depth at best bid = 12 contracts.
2. Post a limit buy at the bid 45.60 and join the back of the queue. Queue_ahead = 12.

3. Walk MBP-10 events (cancels, trades, orders) on the bid and ask side at our price until $t + H = 11:00$. Each such event drains `queue_ahead`.
4. At 10:42 a trade of size 8 prints $\rightarrow$ `queue_ahead` = 4. At 10:51 a trade of size 5 prints $\rightarrow$ `queue_ahead` = -1 $\rightarrow$ filled at 45.60, $t_{\text{fill}} = 10:51$.
5. Monitor the bid through $t + H$: if it touches $45.60 + 0.05 = 45.65$ at some t_{tp} , take-profit close at 45.65. Otherwise force-exit at $t + H = 11:00$ by selling at the then-best bid.
6. In this example the bid never reaches 45.65; force-exit at 11:00 against best bid 45.55. Realised PnL = $(45.55 - 45.60) \cdot \$1,000 - \$2.20 = -\$52.20$ (\$2.20 round-trip fee).
7. If `queue_ahead` > 0 at $t + H$ the order is cancelled, with no trade recorded and no fee charged.
8. Short signals mirror the procedure: post at best ask, monitor for an ask touch, act correspondingly.

7.6 Strategy validation

With $\hat{\Theta}$ frozen, the conviction gate and horizon are not tuned: every (horizon, gate) cell is reported on the full six-month window. To probe regime dependence the window is split into Window 1 (2025-03-02 – 2025-05-30, the high-volatility April tariff-shock) and Window 2 (2025-06-01 – 2025-08-29, the post-tariff calm). The split is a diagnostic, not a selection. The propagator requires no calibration, so both windows lie outside the calibration sample.

7.7 Results

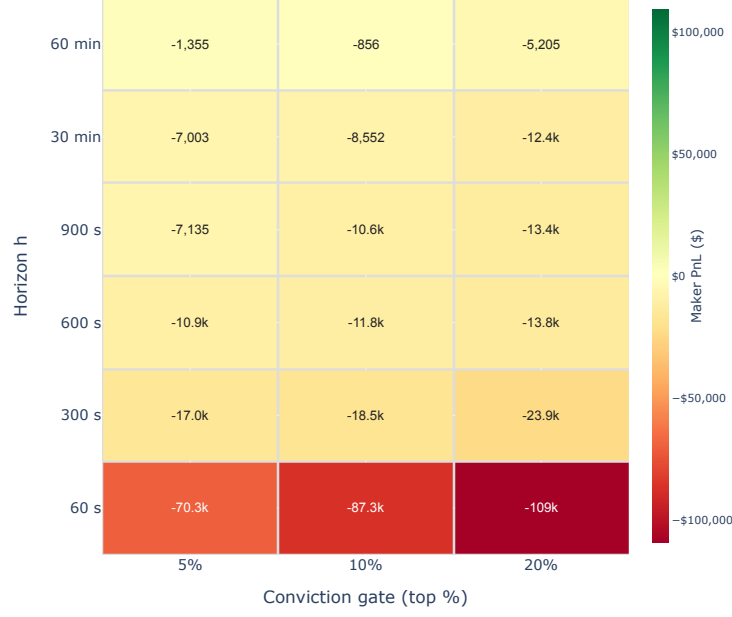


Figure 16: Total PnL over the conviction-gate $\times$ horizon grid.

Under the honest position execution every (horizon, gate) cell loses: the surface in Figure 16 is red across the whole grid, and the eighteen cells of Table 9 sum to $-\$429\text{k}$. The shape of the loss is that of a capped, unstopped, adversely-selected option. The upside is capped at the +1-tick take-profit ($\approx +\$48$ at $h \geq 300\text{ s}$; at $h = 60\text{ s}$ the favourable column is itself negative, $-\$34$ to $-\$86$, because the deadline closes the position before the tick is reached). The downside is not stopped: an adverse fill runs to the $t + H$ deadline, $-\$78$ to $-\$308$ in the top-5% gate. The fills are then drawn from the adverse side, with the adverse-to-favourable fill ratio at or above unity across the grid (strictly above at the bulk of $h \geq 300\text{ s}$ cells).

Table 9: PDV trading performance by horizon and conviction gate, under the cancel-aware honest maker (identical execution to the CatBoost study). Each signal is labelled by whether its realised move ($VX_{t+H} - VX_t$) ran *against* (adv) or *with* (fav) the forecast.

gate	n_{fire}	$P(\text{fill} \mid \text{move})$			$\Delta \text{PnL}(\$)$		PnL (\$)
		adv	fav	ratio	adv	fav	
$h = 60 \text{ s}$							
top 5%	1,329	0.68	0.68	1.00	-78	-86	-70,301
top 10%	2,658	0.70	0.69	1.02	-43	-56	-87,295
top 20%	5,315	0.69	0.68	1.02	-28	-34	-109,009
$h = 300 \text{ s}$							
top 5%	270	0.90	0.70	1.29	-165	+12	-16,958
top 10%	540	0.89	0.71	1.25	-96	+19	-18,528
top 20%	1,079	0.89	0.72	1.23	-71	+24	-23,907
$h = 600 \text{ s}$							
top 5%	138	0.91	0.78	1.16	-206	+42	-10,860
top 10%	275	0.94	0.78	1.20	-129	+46	-11,819
top 20%	550	0.92	0.79	1.16	-95	+47	-13,832
$h = 900 \text{ s}$							
top 5%	94	1.00	0.80	1.24	-228	+45	-7,135
top 10%	187	0.98	0.84	1.16	-182	+48	-10,624
top 20%	373	0.98	0.82	1.18	-131	+46	-13,387
$h = 30 \text{ min}$							
top 5%	50	1.00	0.87	1.15	-308	+35	-7,003
top 10%	99	0.98	0.87	1.12	-234	+44	-8,552
top 20%	197	0.98	0.86	1.14	-177	+48	-12,350
$h = 60 \text{ min}$							
top 5%	25	1.00	1.00	1.00	-169	+52	-1,355
top 10%	50	1.00	0.93	1.07	-117	+57	-856
top 20%	99	0.98	0.90	1.09	-159	+57	-5,205

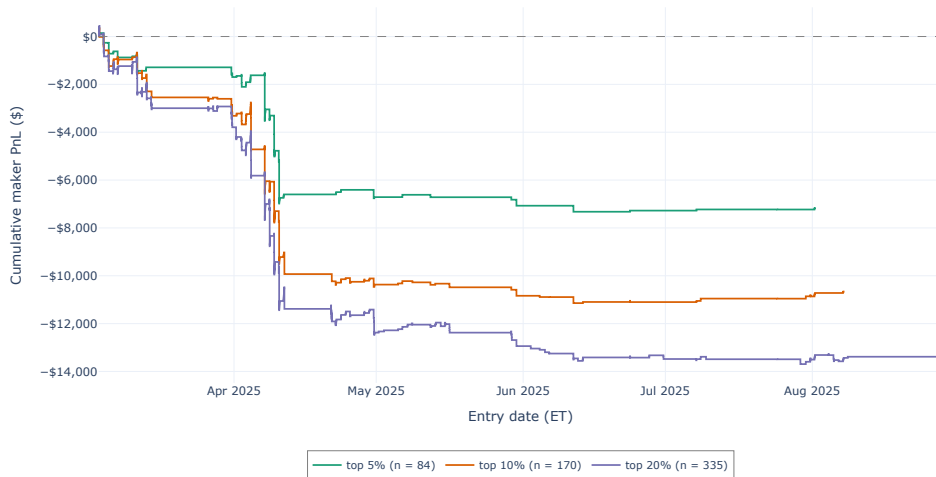


Figure 17: Cumulative maker PnL at $h = 900 \text{ s}$

Regime and channel decomposition

In order to understand the results, we proceed further with the decomposition by window and feature. The cumulative curve at $h = 900$ s (Figure 17) already shows the shape: the loss is accrued in Window 1 and the curve is near-flat through Window 2.

Table 10: Total PnL split by regime window (top 10% gate). The loss concentrates in Window 1, where the strategy fires most and is most often wrong. Window 2 is a near-flat noise.

h	n_{W1}	PnL _{W1} (\$)	n_{W2}	PnL _{W2} (\$)
60 s	1,650	−80,430	143	−6,865
300 s	384	−17,095	38	−1,434
600 s	212	−11,366	24	−453
900 s	153	−10,837	17	+213
30 min	81	−8,728	11	+176
60 min	43	−1,145	5	+289

Table 11: Total PnL (\$) by initiating drift channel. A filled trade is R_1 -initiated when $|\beta_1 \bar{\lambda}_1 \bar{R}_{1,t}| \geq |\frac{\beta_2 \bar{\lambda}_2}{2}(\sigma_t^2 - \bar{R}_{2,t})/\sqrt{\bar{R}_{2,t}}|$ at entry.

Window 1							Window 2						
h	n_{R_1}	Σ_{R_1}	μ_{R_1}	n_{R_2}	Σ_{R_2}	μ_{R_2}	h	n_{R_1}	Σ_{R_1}	μ_{R_1}	n_{R_2}	Σ_{R_2}	μ_{R_2}
60 s	331	+2,572	+8	1,319	−83,002	−63	60 s	0	—	—	143	−6,865	−48
300 s	45	+1,951	+43	339	−19,046	−56	300 s	0	—	—	38	−1,434	−38
600 s	17	−37	−2	195	−11,329	−58	600 s	0	—	—	24	−453	−19
900 s	13	−779	−60	140	−10,058	−72	900 s	0	—	—	17	+213	+13
30 min	5	+289	+58	76	−9,017	−119	30 min	0	—	—	11	+176	+16
60 min	4	+241	+60	39	−1,386	−36	60 min	0	—	—	5	+289	+58

1. Only the mean-reversion drift channel monetises (Table 11). The R_1 -drift $-\beta_1 \bar{\lambda}_1 \bar{R}_{1,t}$ erases the leverage skew $\beta_1 \bar{R}_{1,t}$ off the level $\sigma = \beta_0 + \beta_1 \bar{R}_{1,t} + \beta_2 \sqrt{\bar{R}_{2,t}}$, pulling σ back toward the variance-anchored baseline $\beta_0 + \beta_2 \sqrt{\bar{R}_{2,t}}$; its fixed point is $\bar{R}_{1,t} = 0$. It bets correctly on this mean reversion in the high-vol regime (positive μ_{R_1} at most horizons in Window 1), but its scale is too small to clear costs in the calm regime, and in Window 2 it never fires at all (Table 10). The R_2 -drift channel, by contrast, is empirically near-zero-to-negative noise: it initiates most of the fills yet books a negative mean per trade at every horizon (−\$54 to −\$102 per trade), so the variance-gap term exists in the formula but does not pay.

Forecasting performance

Table 12: Quantiles of the forecast magnitude $|f_t(H)|$ by horizon. For comparison the typical VX bid–ask spread is 0.05.

h (s)	p_{50}	p_{75}	p_{90}	p_{95}	p_{99}	n
60	0.0001	0.0001	0.0003	0.0006	0.0027	26,573
300	0.0003	0.0007	0.0017	0.0033	0.0161	5,394
600	0.0007	0.0016	0.0037	0.0070	0.0291	2,746
900	0.0010	0.0023	0.0057	0.0111	0.0576	1,864

Table 13: Forecasting skill by horizon. $n_{\neq 0}$ is the directional-accuracy denominator.

h (s)	MAE ₁	MAE ₀	Dir. acc.	DM (p)	$n_{\neq 0}$	n
60	0.032	0.032	0.490	+1.13 (0.258)	10,574	26,573
300	0.073	0.073	0.485	+0.97 (0.332)	3,357	5,394
600	0.105	0.105	0.479	+1.03 (0.305)	1,951	2,746
900	0.132	0.131	0.500	+0.75 (0.454)	1,396	1,864

1. No directional skill at any horizon: directional accuracy sits at 0.48–0.50 (Table 13), indistinguishable from the coin toss, and the gap from 0.5 is on the wrong side at three of the four horizons.
2. As a value forecast it does not beat a no-change benchmark: MAE₁ $\approx$ MAE₀ at every horizon and Diebold–Mariano never rejects ($p \geq 0.26$, signed DM mildly positive, Table 13), so the forecast’s squared error is the same as or larger than that of predicting no move.
3. The forecast is deeply sub-tick: even its p_{99} magnitude stays below the 0.05 VX tick at the short horizons (Table 12), so a correct sign could not clear the spread.

The PDV-SDE forecast is empty on VX intraday: it predicts neither the sign (Table 13) nor the scale (Table 12) of the next move, and trades at a loss at every horizon and conviction gate.

7.8 Summary

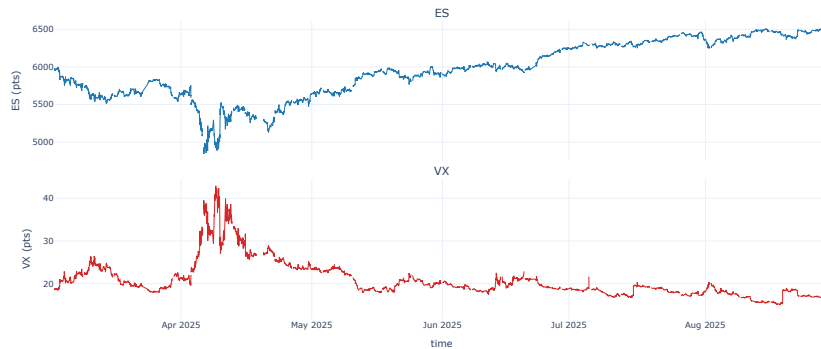


Figure 18: ES and VX

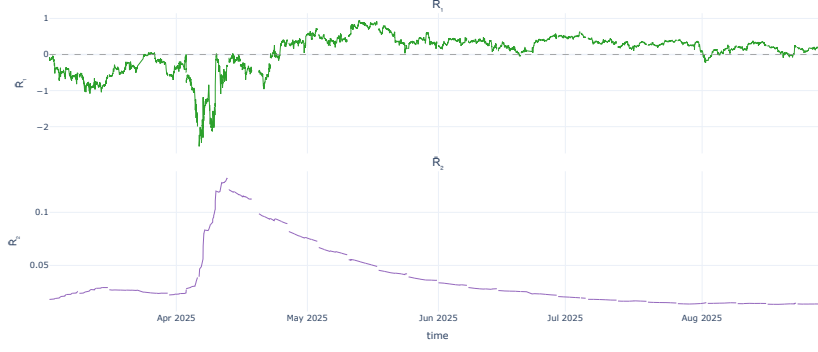


Figure 19: The state quantities $\bar{R}_{1,t}$ and $\bar{R}_{2,t}$



Figure 20: $R_1\text{-drift} = -\beta_1 \bar{\lambda}_1 \bar{R}_{1,t}$ and $R_2\text{-drift} = \frac{\beta_2 \bar{\lambda}_2}{2} (\sigma_t^2 - \bar{R}_{2,t}) / \sqrt{\bar{R}_{2,t}}$

1. Pure Guyon's PDV is not enough for stable PnL growth across regimes. The drift channels R_1 and R_2 capture mean reversion and volatility clustering, but the leverage effect lives entirely in the diffusion $\beta_1 \bar{\lambda}_1 \sigma_t dW_t = \beta_1 \bar{\lambda}_1 \frac{dS_t}{S_t}$. The model captures that term correctly; the strategy does not, because it trades on $\mathbb{E}[d\sigma_t | \mathcal{F}_t]$ and the diffusion is a martingale increment that vanishes under conditioning. The strategy is therefore blind to the leverage shock by construction, not because the model omits it (Figure 18 shows the ES–VX co-movement it cannot trade), and directional accuracy stays at chance.
2. R_1 pays off in the high-vol regime, where it correctly bets on the mean reversion of volatility, but rarely fires in the low-vol regime and is too small there to clear costs (Figures 19–20).
3. R_2 updates cannot separate signal from noise at one-minute bars (Figure 19): the variance-gap drift is dominated by single-bar return noise, so its per-trade PnL is negative. Consequently, the R_2 channel is the sink that carries the bulk of the loss.
4. The scale of $\mu \cdot dt$ is lower than the bid–ask spread. The model enters when the forecast is sub-spread, so even a correctly-signed move cannot pay for the round trip.

8 Forecasting. CatBoost Approach

The main idea is to attempt catching changes in VX futures through the microstructure of the order book^{[17][18]}. We introduce a machine-learning approach^[21] as it requires no model set-up and is able to catch non-linearities within the features.

8.1 Features

The forecaster is fed by a set of 15 features split into 4 blocks.

The six order-book features below are computed once on the ES book and once on the VX book; each is explained here^[18].

ES block

1. Order-book imbalance

$$\text{imbalance}_L = \frac{\sum_{i=1}^L q_i^{\text{bid}} - \sum_{i=1}^L q_i^{\text{ask}}}{\sum_{i=1}^L q_i^{\text{bid}} + \sum_{i=1}^L q_i^{\text{ask}}}, \quad L \in \{1, 3, 5\}$$

Net resting size: a positive value means more buyers than sellers, which tends to precede a short upward move.

2. Microprice deviation

$$\Delta_t = \frac{\mu_t - m_t}{ES_{\text{tick}}} \quad \text{computed by}$$

$$\mu_t = \frac{\text{ask}_t q_1^{\text{bid}} + \text{bid}_t q_1^{\text{ask}}}{q_1^{\text{bid}} + q_1^{\text{ask}}}$$

$$m_t = \frac{\text{ask}_t + \text{bid}_t}{2}$$

The size-weighted fair price minus the mid: it leans toward the heavier side and anticipates where the mid will settle next.

3. Spread

$$\text{ask}_t - \text{bid}_t$$

Cost of immediacy and a liquidity gauge: a widening spread flags uncertainty and raises the bar a forecast must clear to trade.

4. Log-returns

$$\log(m_t / m_{t-\Delta}), \quad \Delta \in \{5s, 10s, 20s\}$$

Recent price drift over three windows: captures short-horizon momentum or reversal in the mid.

5. Realized volatility

$$\sqrt{\sum_{(t-\Delta, t]} r_s^2}, \quad \Delta \in \{20s, 40s, 60s\}$$

Local activity level: high recent volatility makes a larger next move more likely and scales the others.

6. MACD

$$\text{EMA}_{10s}(m_t) - \text{EMA}_{20s}(m_t)$$

Fast minus slow moving average: a compact trend signal whose sign marks the prevailing short-term direction.

VX block

The same six features computed on the VX book itself. VX is the traded instrument, so its own imbalance and microprice are the most direct read on the next VX move, while ES adds additional features source.

PDV block

The two drift channels of the calibrated PDV SDE from the previous section, evaluated intraday: they inject the model's view of where volatility is heading.

1. R_1 drift

$$R_1\text{-drift} = -\beta_1 \bar{\lambda}_1 \bar{R}_{1,t}$$

The trend channel: a recent run pulls volatility back, so its sign and size forecast the next move in σ .

2. R_2 drift

$$R_2\text{-drift} = \frac{\beta_2 \bar{\lambda}_2}{2} \cdot \frac{\sigma_t^2 - \bar{R}_{2,t}}{\sqrt{R_{2,t}}}$$

The variance channel: how far current variance sits above its path-dependent anchor, the part the trend channel does not explain.

Time block

1. Session progress

$$\frac{\text{seconds since 09:30 ET}}{23400}$$

Fraction of the trading day elapsed: lets the model learn the intraday seasonality of volatility and liquidity (open and close are most active).

8.2 Model used

We use a CatBoost model on the features above^[22]. The idea is simple. Start with a simple mean prediction and proceed by growing decision trees over the set of features. Each new tree corrects the loss of the previous tree.

The target is the change in the VX-futures mid over a horizon $h \in \{40, 60, 300, 600, 900\}$.

$$\Delta \text{VX}_t^h = m_{t+h} - m_t, \quad m_t = \frac{\text{ask}_t + \text{bid}_t}{2},$$

Under L_2 loss, CatBoost estimates the conditional mean $\mathbb{E}(\Delta \text{VX}_t^h \mid \mathcal{F}_t)$, the empirical analog of Guyon's $\mathbb{E}(d\sigma_t \mid \mathcal{F}_t)$ ^[7]. This target allows a direct comparison with the parametric SDE forecast of the previous section.

8.3 Calibration and Validation

Training, validation and testing use a settlement-aware walk-forward over the full six-month sample from 2025-03-02 to 2025-08-29:

1. Each fold is 7 consecutive training days, 2 validation days and 1 test day, sliding forward one day at a time.
2. The validation days select the tree depth with the lowest validation MAE over $\{2, 4, 6\}$ and early-stop the boosting after 50 rounds.
3. Any fold within ± 3 days of a VX settlement is rejected.
4. Out-of-sample predictions are then pooled across folds.

8.4 Results

Feature importance

Table 14: Walk-forward feature importance by block (%), fold-averaged. PDV is split into its two drifts.

h (s)	ES	VX	R_1 -drift	R_2 -drift	Time
40	8.2	73.5	3.0	12.4	2.9
60	5.4	73.9	4.7	12.7	3.3
300	8.4	30.5	45.1	10.3	5.7
600	7.1	23.9	53.7	8.7	6.6
900	6.3	15.4	63.3	8.1	6.9

Table 14 reads as follows.

1. At 40–60s the VX book holds $\sim 74\%$ of importance.
2. Weight shifts to the R_1 -drift and intraday time as h grows; the VX book falls below 30% by 600s.
3. The two PDV drifts cross over: R_2 -drift leads at short h , R_1 -drift dominates at long h , confirming the finding of the SDE approach.
4. ES stays at 5–8% throughout.

Trading performance

A signal is executed only if it falls in the top $p \in \{5, 10, 20\}\%$ of blocks by predicted magnitude $|\widehat{\Delta VX}|$. Figure 21 maps the resulting cumulative PnL across the gate-by-horizon grid, and Figure 22 traces it through time at the one profitable horizon.

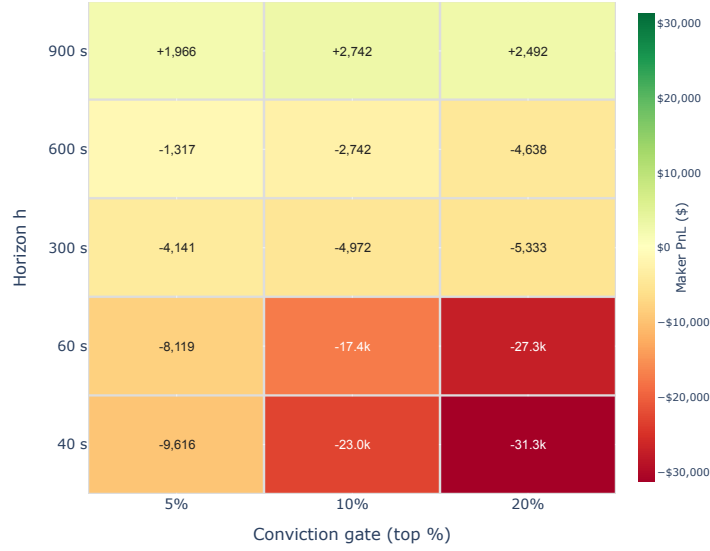


Figure 21: Cumulative PnL (\$) over the conviction-gate (5/10/20%) by horizon grid

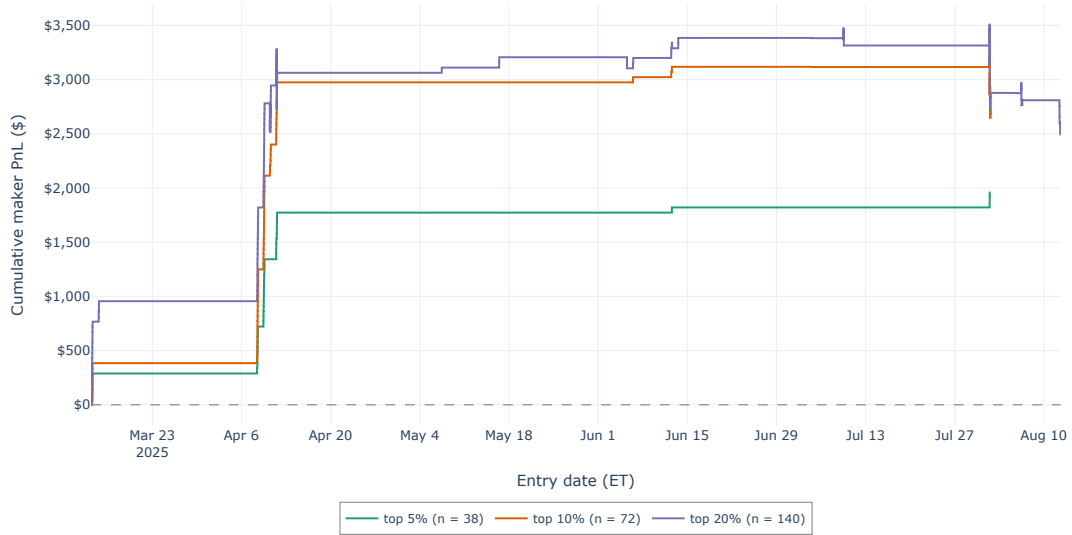


Figure 22: Cumulative PnL at the horizon ($h = 900s$) per conviction gate (5/10/20%).

Table 15: Trading performance by horizon and conviction gate. Each signal is labelled by whether its realised move ($VX_{t+h} - VX_t$) ran *against* (adv) or *with* (fav) the forecast direction.

gate	n_{fire}	$P(\text{fill} \mid \text{move})$			$\Delta\text{PnL}(\$)$		PnL (\$)
		adv	fav	ratio	adv	fav	
$h = 40 \text{ s}$							
top 5%	545	0.96	0.66	1.46	−97	+52	−9,616
top 10%	1,455	0.95	0.63	1.50	−88	+51	−23,032
top 20%	2,219	0.96	0.51	1.87	−85	+50	−31,349
$h = 60 \text{ s}$							
top 5%	393	0.98	0.70	1.40	−100	+54	−8,119
top 10%	788	0.98	0.65	1.52	−102	+51	−17,412
top 20%	1,606	0.96	0.51	1.88	−92	+51	−27,302
$h = 300 \text{ s}$							
top 5%	102	0.98	0.71	1.38	−103	+50	−4,141
top 10%	226	0.99	0.76	1.31	−95	+49	−4,972
top 20%	406	0.99	0.73	1.36	−78	+50	−5,333
$h = 600 \text{ s}$							
top 5%	55	0.96	0.96	1.01	−102	+50	−1,317
top 10%	117	0.98	0.89	1.10	−105	+49	−2,742
top 20%	217	0.99	0.85	1.17	−96	+51	−4,638
$h = 900 \text{ s}$							
top 5%	38	1.00	1.00	1.00	+48	+54	+1,966
top 10%	75	1.00	0.93	1.07	+18	+53	+2,742
top 20%	151	1.00	0.91	1.10	−16	+51	+2,492

Table 15 decomposes every gate by fill direction.

1. The strategy is a capped, unstopped option that is adversely selected. It fills $1.5\times$ more into losing moves ($P(\text{fill} \mid \text{adv})/P(\text{fill} \mid \text{fav}) = 1.50$ at 40s) and is paid asymmetrically: a favourable-move fill books the capped +1 tick ($\approx +\$50$), while an adverse one runs unstopped to the deadline ($-\$88$). At 40s the top-10% gate fires on over 1,151 fills and loses $-\$23\text{k}$, the worst cell in the grid.
2. As h grows, adverse selection fades (ratio from 1.50 to 1.07) and the scale of the predicted move clears the spread, so per-fill PnL rises from $-\$20$ to $+\$38$. At 900s it is pure spread capture: even adverse-move fills average $+\$18$, on just 38–151 trades, and the cell turns positive ($+\$2.7\text{k}$).

In short, the strategy monetises time, direction and the bid–ask spread, not the magnitude of the forecast. At the horizon where the signal is strongest the execution economics are the worst, because that is exactly where the order book picks off the resting quote (Figure 21).

Forecasting performance

Table 16: Quantiles of the CatBoost forecast magnitude by horizon. For comparison the VX bid–ask spread is 0.05

h (s)	p_{50}	p_{75}	p_{90}	p_{95}	p_{99}	n
40	0.0005	0.0013	0.0033	0.0144	0.0553	10,721
60	0.0006	0.0016	0.0037	0.0053	0.0514	7,717
300	0.0028	0.0077	0.0274	0.0419	0.2281	1,984
600	0.0047	0.0142	0.0492	0.0871	0.4402	1,061
900	0.0089	0.0251	0.0673	0.1316	0.3580	732

Table 17: Forecasting skill by horizon.

h (s)	MAE_1	MAE_0	Dir. acc.		DM (p)	$n_{\neq 0}$	n
			real	perm.			
40	0.044	0.043	0.589	0.511	+1.72 (0.085)	4,620	10,721
60	0.053	0.052	0.577	0.510	+2.80 (0.005)	3,684	7,717
300	0.111	0.107	0.549	0.517	+2.52 (0.012)	1,315	1,984
600	0.163	0.155	0.510	0.501	+2.22 (0.027)	777	1,061
900	0.184	0.180	0.521	0.504	+0.97 (0.330)	560	732

Table 17 reports the skill and Table 16 the size of the forecasts.

1. A directional edge fades with h : dir. acc. 0.589 at 40s and 0.577 at 60s, against a shuffled-label baseline of 0.51. By 600s it is back to that baseline. Permuting the training labels collapses every all-sample accuracy to 0.50–0.52, so the edge is leakage-free, not a fit to noise.
2. DM exceeds zero at every horizon and rejects at 60–600s: the model’s squared error is *larger* than the no-change benchmark. The model continuously bets on small changes, but since about 40% of the sample has no change at all, betting on a move does not pay off on average. The skill is one-sided, in the sign, not the magnitude.
3. The forecasts themselves are tiny: at 40s the median $|\widehat{\Delta VX}|$ is 0.0005 and even the 95th percentile (0.0144) is under a third of the 0.05 spread (Table 16), which is why the sign edge never clears the round trip.
4. The skill lives where the VX book dominates importance (Table 14) and vanishes where the PDV drifts take over: the directional edge is an order-book effect, not a PDV one.

The order book carries a faint, leakage-free directional edge at the sub-minute scale, one the PDV drifts do not have. Yet it is too weak to clear the bid–ask and is adversely selected under any realistic fill: the model can forecast the sign of the next VX move, but it cannot be traded.

9 Conclusion

9.1 Main results

Main results of this work can be summarised by the following Q&A.

Q: Is pure PDV acceptable for VX trading?

No. The drift carries no directional skill at any horizon and never beats a no-change benchmark (Table 13). Its forecast is sub-spread against a 0.05 tick. Sign-blind and too small to trade.

Q: Are the PDV features usable?

Depends on the feature. R_1 captures mean reversion of volatility, but only in the high-vol Window 1, and even there it barely breaks even. In the calm Window 2 it never fires. R_2 at one-minute updates is noise: it carries the entire trading loss, $-\$54$ to $-\$102$ per trade (Table 11). Handed to the CatBoost model as features, the R_1 -drift dominates importance but still yields little skill. For R_2 either a slower clock or a noise-filtering algorithm is required.

Is the order-book approach viable?

Yes, with two corrections. Its edge lives at the sub-minute scale, where adverse selection is worst (Table 17). The signed-move forecast was also fit under L_2 loss only to match Guyon’s value forecast. That is the wrong objective: the forecast has sign skill but is magnitude-poor. A direct move classifier that targets the change by tick would serve better.

Is the strategy logic viable?

No. It is a capped, unstopped option entered on a sub-spread forecast: a favourable fill books one tick, an adverse fill runs to the deadline, and it fills more often into the adverse move. The fix is a market-making frame: two-sided quotes with inventory and adverse-selection control driven by the move classifier above that targets spread capture.

9.2 Additional results

Additional results of this work may be branched into two separate results.

Theoretical derivation.

The TSPL path-dependent volatility model can be rewritten as an infinite-dimensional Markovian model with a continuity of tail parameters. Projecting the infinite-dimensional model onto two-exponential forms collapses it to a four-factor Markovian model. Guyon’s findings claim that 2 λ ’s are sufficient to capture infinite dimensionality. The infinite-dimensional model also explains the vol-of-vol parameter: δ_1 sets the attention horizon, and a large δ_1 smooths attention across past returns and damps vol-of-vol (Remark 5.10).

Fitting finding.

R_2 can saturate. The model’s largest systematic residuals come from interventions where policy outpaces the slow R_2 decay (2008, March 2020). Adding VIX_{t-1} and updating via steps restores non-saturation (§6.2).

10 Future work

The findings open two branches of future work, set by the question one asks: "How can the PDV model be improved?", or "Is stable PnL growth the goal?"

10.1 Improving the PDV model

The fitting analysis (§6.2) points to three additions.

1. **Return features.** A large part of the residual variance traces to foreign-market shocks, so adding FX returns or European and Asian index returns should help. A further step is a synthetic “world index” that blends SPX, NKY and other indices into the return driver. This doesn’t directly holds with the VIX being derived from SPX alone, but the residuals tell a different story.
2. **Macro features.** The magnitude and direction of central-bank policy are not directly predictable, yet they are predictable in the mean, so a scheduled-event calendar (FOMC, NFP, CPI) adds a pricing source. The cost is the elegance of the original model.
3. **Intraday updates.** Adding intraday returns improves the metrics, but at too small scale the R_2 updates turn unreliable. This calls for noise filtering, or for replacing the realized variance with an implied estimate^[23].

10.2 Stable PnL growth

Taking the market directly does not work: the signal is adversely selected, and the L_2 loss on returns is the wrong objective. The natural replacement is a direct classifier of the next move paired with a probability score^[21], executed not by crossing the spread but as a two-sided market-making frame with inventory and adverse-selection control^[20]. Adverse selection does not vanish, but spread capture is there to pay.

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